

Introduction to Multimodal AI

Course Title: Multimodal AI

Course Code: AAI3031-01

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1. Course Information

- **3 Credits** [Flipped Learning (1 hour online/recorded, 2 hours offline)]

Lectures: 3 hours weekly

- Room: International Hall Multimedia Lecture Room (**9B217**)
 - Thursday 15:00 to 15:50 PM
 - Thursday 16:00 to 16:50 PM
 - Total: 1 Hour, 40 min.
- **Type:** Theory/practical/invited speeches/Recorded/online/in-person
- **Attendance**
 - “SKKU-Attend” for attendance, absence and late rules
 - Default setting for the time being
 - Zoom/Manual



성균관대학교 전자출결
Sungkyunkwan University

Uninstall

Open

2. Course Materials

- **PDF/PPT of slides**
- The slides will be available on i-Campus of SKKU
- Sometimes the slides will be updated later.
- Recorded videos/lectures will be available on i-Campus (Lecture Contents)
- Some more references and links for lectures notes may be given time to time
- Tools

Depending on the assigned project, you will have choice to select language/toolkit/frameworks (s):

- Python, C++
- MATLAB, Julia
- PyTorch, TorchVision, OpenCV etc.

3. Students' Assessment

- Attendance: 10%
- Mid-term exam: 20%
- Final-term exam: 40%
- Practice Assignment (PA)/Quizzes/Homework etc.: 15%
- Presentation: 15%
- **Project**
 - Multimodal AI and your major related project completion
 - Survey/Research articles
- **Subjective**
 - Short-answer essay
 - Extended-response essay
 - Problem solving
- **Cheating/Copy Paste Policy**
 - Automatic **F** for both



4. Contact Instructions

E-mail (Preferred)

- khanmuhammad@g.skku.edu (preferred)
- khanmuhammad@skku.edu
- Information needed (in English)
 - i. Subject of the Email
 - ii. Student Name
 - iii. Student ID
 - iv. Email's Content/Query/Question/Feedback/Suggestions etc.

Office

- International Hall, Room # 90326
- A prior appointment through e-mail is required (extremely serious matters)
- Online Zoom/Google Meet meetings can be arranged with flexibility.

TA

- Wajahat, Master Student (wajahatullah@g.skku.edu)

5. Course Rules/Regulations

- Assignment/Homework via e-mail is not encouraged (exceptional).
- [Assessment rubrics](#) will be followed to evaluate assignments/projects and exams (depending on the need).
- University/Department rules and regulations must be followed.
- More: Will be explained from time to time

6. Professor's Bio



Khan Muhammad (Senior Member, IEEE)

- Associate Professor (Tenure-Track), SKKU, Korea
- khanmuhammad@skku.edu
- +82-10-4831-2104
- Editorial Board Member, 7 Journals
- Area/Associate Editor, 7 Journals
- Top 2% Researcher (Stanford Uni. List), 2020-present
- #2 in Korea (Best CS Scientists 2026, Research.com)
- HCR (top 1%), 2021-2024

Publication Record

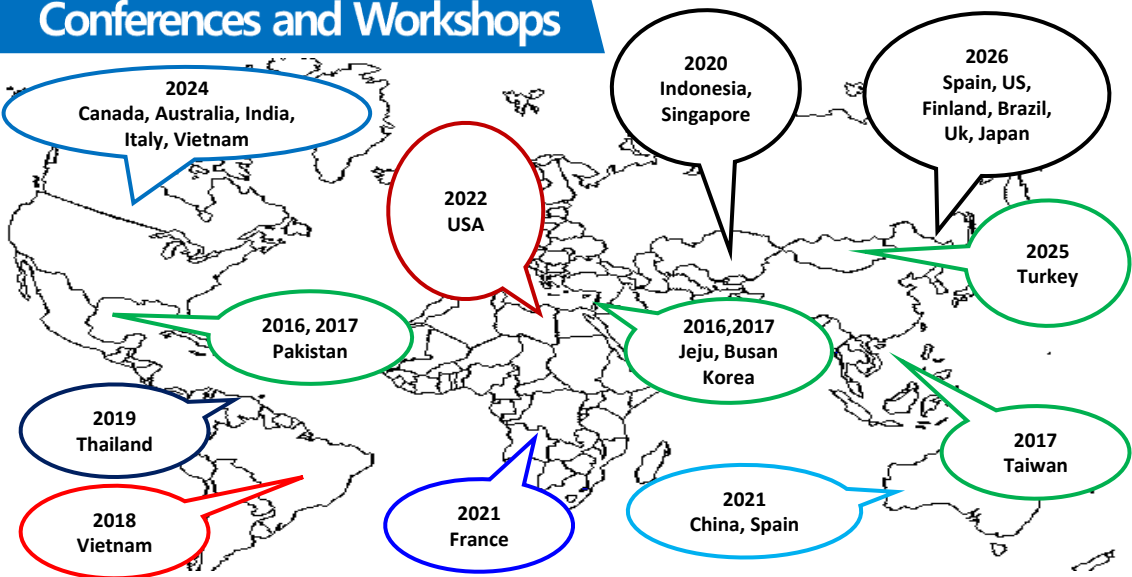
Over 300 articles/papers

As lead-author (1st or corresponding author)	180+
As co-author	90+
Total citations (Google Scholar)	<u>32230+</u> (H-Index: 93)
Impact factor	1750+
Patents Record (Korean, US mainly)	15
Participation in Research Projects	15
Professional and Research Experience	10+

Research Interests

- ① **Computer Vision**
 - Activity Recognition (Normal, Child Hazards, Aggression, Bullying)
 - Image Enhancement (Low-light, Dehazing, Denoising)
 - Person Re-identification
- ② **Large Language Models (LLMs)**
 - Sentiment Analysis
 - Hallucinations
- ③ **Intelligent Surveillance**
 - Fire/smoke detection, segmentation and scene analysis
 - Unexpected Road Hazards Detection and Analysis

Conferences and Workshops



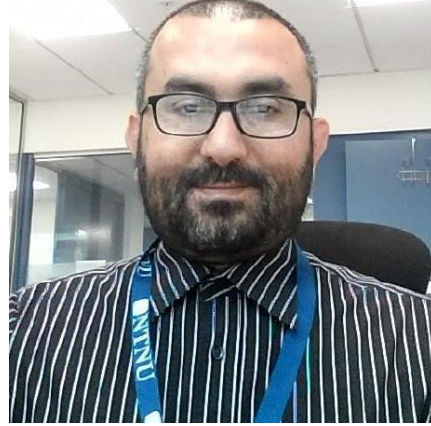
6. Research Team (Major Collaborations)



6. Research Team



Prof. [Sung Wook Baik](#)
(Supervisor, Sejong University, South Korea) [March 2015-Feb. 2019]



Prof. [Muhammad Sajjad](#) (Academic Mentor)
NTNU, Norway and ICP, Pakistan)



Prof. [Irfan Mehmood](#) (Collaborator)
University of Bradford, UK)



Prof. [Jamil Ahmad](#) (BS Supervisor & Collaborator)
Islamia College Peshawar, Pakistan)
MBZUAI, UAE



Abdul Malik Badshah (Master) [Industry]
PhD Fellow, University of Missouri–Kansas City, USA



Noor Ullah (Master) [Industry-Dabeoo, Korea]



Nasir Rahim (Master) [Industry + SKKU]
PDF at University of Michigan, USA

6. Research Team (Alumni)



Dr. Amin Ullah
(Joint Master & Ph.D, SJU, South Korea)
[PDF at Oregon State University, USA]
[Senior AI/ML Engineer at Boeing Research & Technology Software, USA]



Dr. Ijaz Ul Haq
(Joint Master & Ph.D.) (2017-2022), SJU
[PDF at Oakland/Michigan University, USA]



Dr. Tanveer Hussain
(Joint Master & Ph.D.) (2017-2022), SJU
[PDF at University of Leeds, UK]
Lecturer at Edge Hill University, UK]



Dr. Salman Khan
(Master, SJU, South Korea, PhD, Orford-Brookes University, UK)
[PDF at Oxford Brookes University, UK]



Dr. Fath U Min Ullah (Joint Master & Ph.D.)
(2017-2023), SJU, South Korea
[PDF at University of Sheffield, UK]
Lecturer, University of Central Lancashire, UK



Dr. Nasir Rahim (Master, SJU)
[Industry + SKKU (PhD)]
PDF, University of Michigan, USA

6. Research Team (Mix)



Dr. Waseem Ullah (PhD,
2019-2024, SJU)
PDF, MBZUAI, UAE



Dr. Samee Ullah (Ph.D.) (2019-2023)
SJU, South Korea
[PDF, Khalifa Uni. of Science and
Technology, UAE



Zulfiqar Ahmad Khan
(Joint Master & PhD)
(Feb 2019-Feb 2026)
PDF, Umea Uni. Sweden



Siraj, (Ph.D.)
ICP, Pakistan



Muhammad Irfan, MS, SJU PhD,
Oakland Uni. USA



Hayat Ullah, MS, SJU
PhD, Kansas State Uni. USA



Umair Haroon (MS, SJU, 2020-2021)
PhD, University of Barcelona, Spain

6. Research Team (VIS2KNOW Lab)



Farhan Ahmed
(PhD Fellow, SKKU) (March 2021-now)



Fatma Eid
(PhD Fellow, SKKU), (Fall 2023-now)



Zubayda
(PhD Fellow, SKKU), (Spring 2024-now)



Shehzad Ali
(Master-PhD, SKKU), (Spring 2024-now)



MD Tanvir Islam
(Master, SKKU)
(Feb 2023-Aug 2025), PhD, KNU



Bilol
(Master, SKKU)
(Sept 2025-Aug 2027)



Wajahat (TA)
(Master, SKKU)
(March 2026-Feb 2028)



Glody
(Master, SKKU)
(March 2026-Feb 2028)



JeasonHan
(Master, SKKU)



Asad Ullah
(Master, SKKU)
(Sept 2026– Aug 2028)



Muhammad Farooq
(Master, SKKU)
(Sept 2026– Aug 2028)



SeoYoung
(Undergraduate, SKKU)



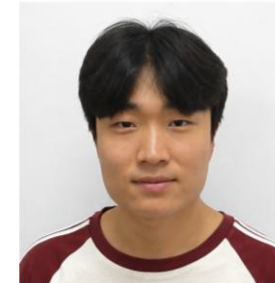
Haong Kim
(Undergraduate, SKKU)



Byeon Jihwan
(Undergraduate, SKKU)



Nisrine Khnifass
(Exchange student, SKKU)



Kim Jae-Yoon
(Undergraduate, SKKU)



Kim Dong-Yoon
(Undergraduate, SKKU)

7. Research Topics

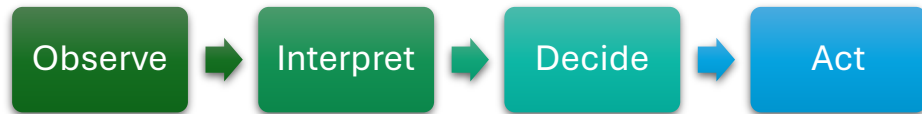


8. Artificial Intelligence (AI)

8 (i): Intelligence Starts with the World

A reasonable question to think on it.

How Do We Make Sense of the World?



Example: A Human Crossing a Road

1. Observe

See cars, pedestrians, traffic lights

2. Interpret

“Is it safe to cross?”

3. Decide

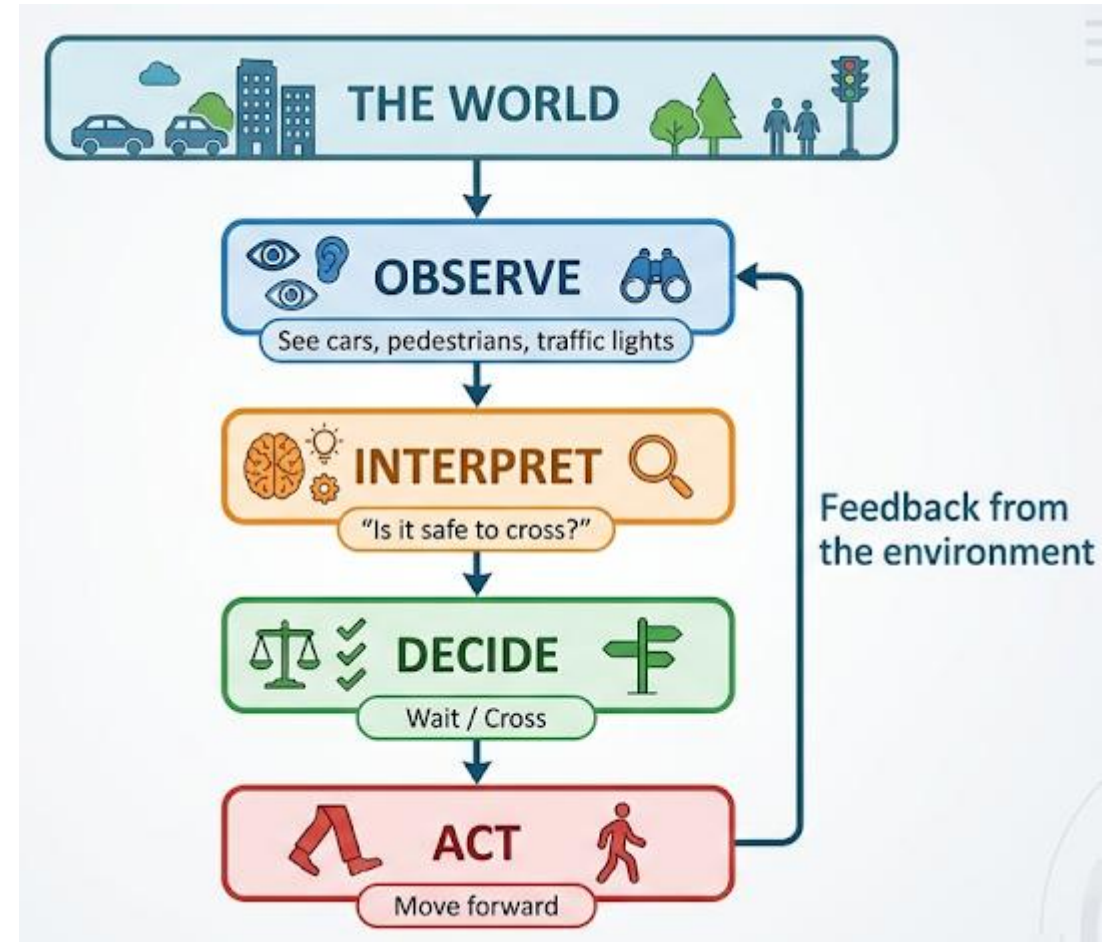
Wait / Cross

4. Act

Move forward

A person can:

- Recognize a familiar face
- Understand spoken language
- Detect danger
- Learn from experience
- Choose an action



- **Intelligence begins with using information to understand a situation and make useful decisions.**

8 (ii): What Is Artificial Intelligence?

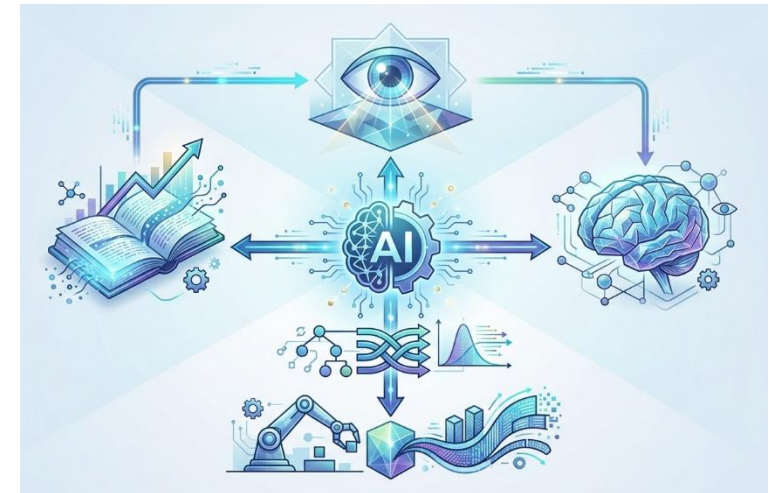
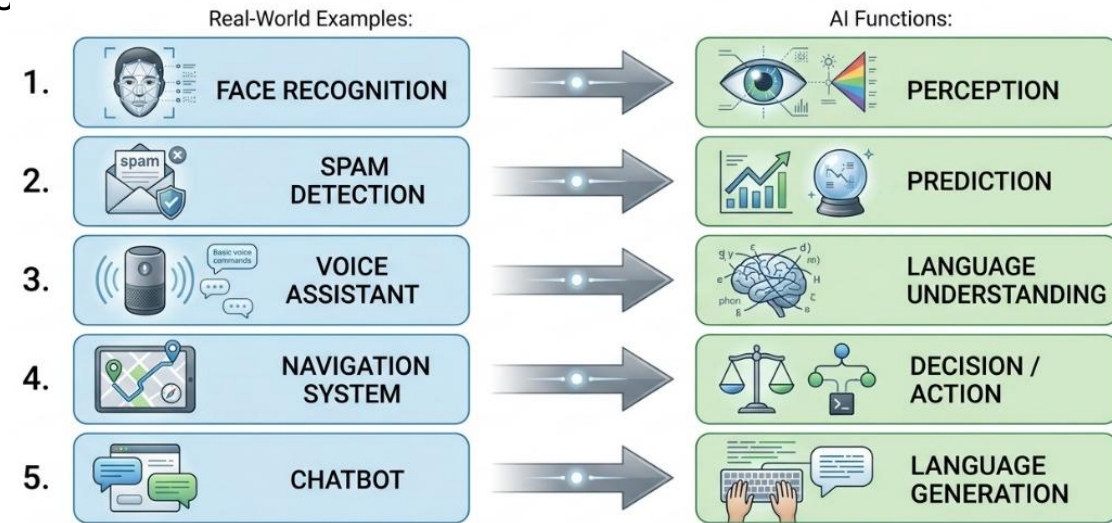
Definition:

Artificial Intelligence (AI) is the field of building computational systems that can perform tasks that normally require aspects of human intelligence.

AI can help machines:

- **Perceive**
See and hear the environment
- **Learn**
Learn patterns from data
- **Understand**
Interpret language and information
- **Reason / predict**
Use information to reach conclusions or predictions
- **Act / generate**
Choose actions or produce new content

AI performs specific tasks that involve capabilities associated with intelligence. The example for it is given in the diagram.



8 (iii): Intelligence in Machines: Rules vs Learning

How Do We Teach a Machine to Solve a Problem?

We can explicitly program the solution OR we can give the machine examples and let it learn the pattern.

1:Explicit Rules

“Tell me what to do?”

2:Machine Learning

“Show me examples, So I can learn pattern”

Example:1

IF message contains suspicious pattern

↓

SPAM

ELSE

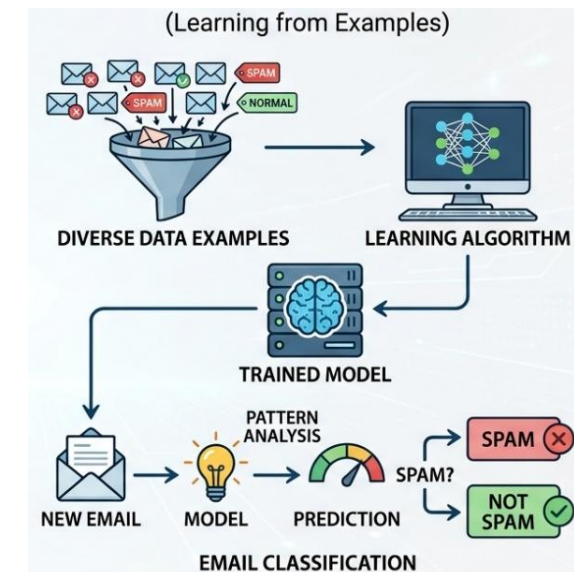
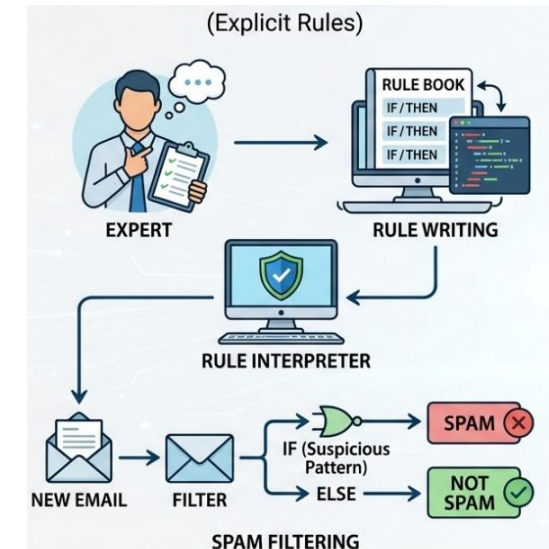
↓

NOT SPAM

Example: 2

Many spam + normal emails

- Learning
- Learned model
- New email → Spam / Not Spam

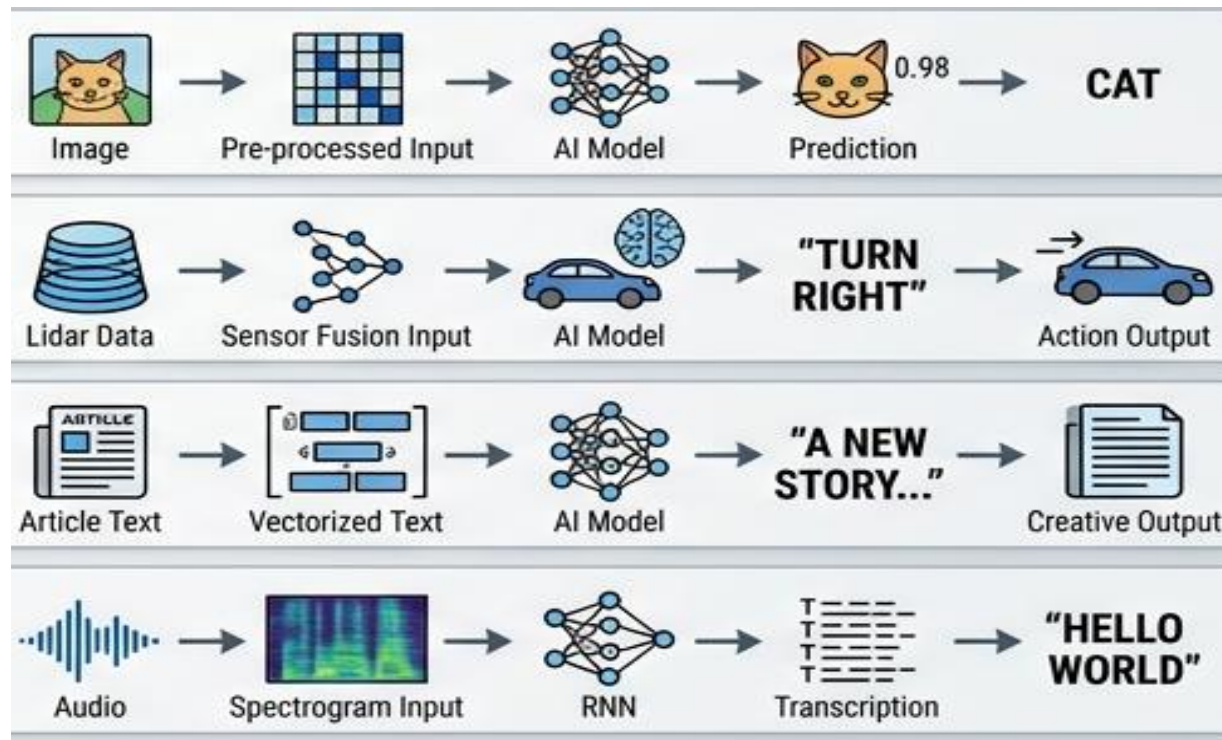


8 (iv): The Basic Idea of an AI System

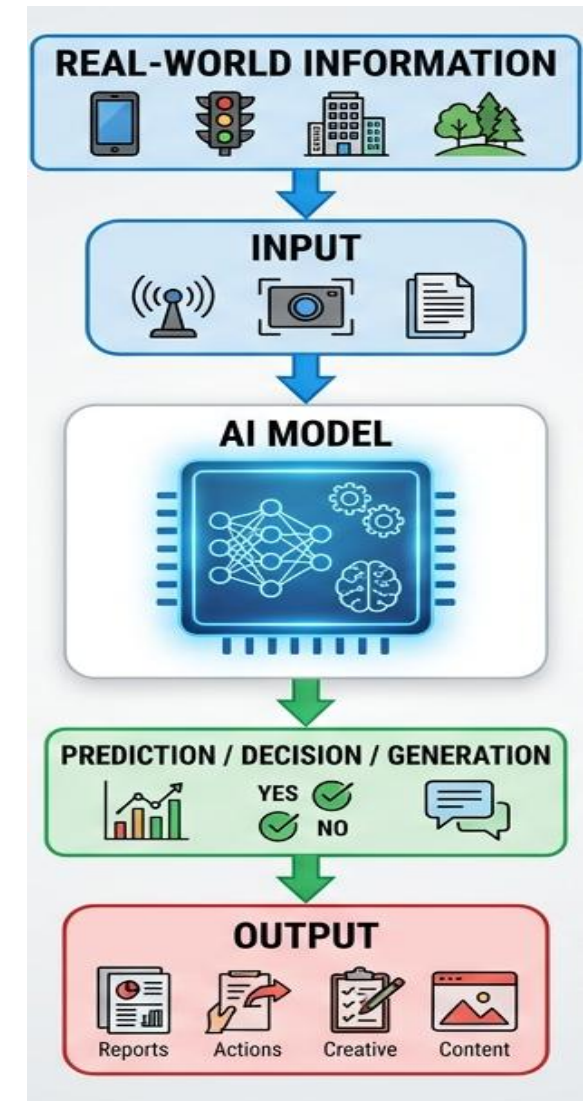
Basic Idea



Some Visual Examples



➤ An AI system transforms input information into a useful output.



9. Multimodal AI

9 (i): The Real-World Problem

One event. Many signals.

- Imagine an autonomous vehicle approaching a busy crossing at night.
- A single real-world situation generates multiple, complementary (different but useful together) sources of information.

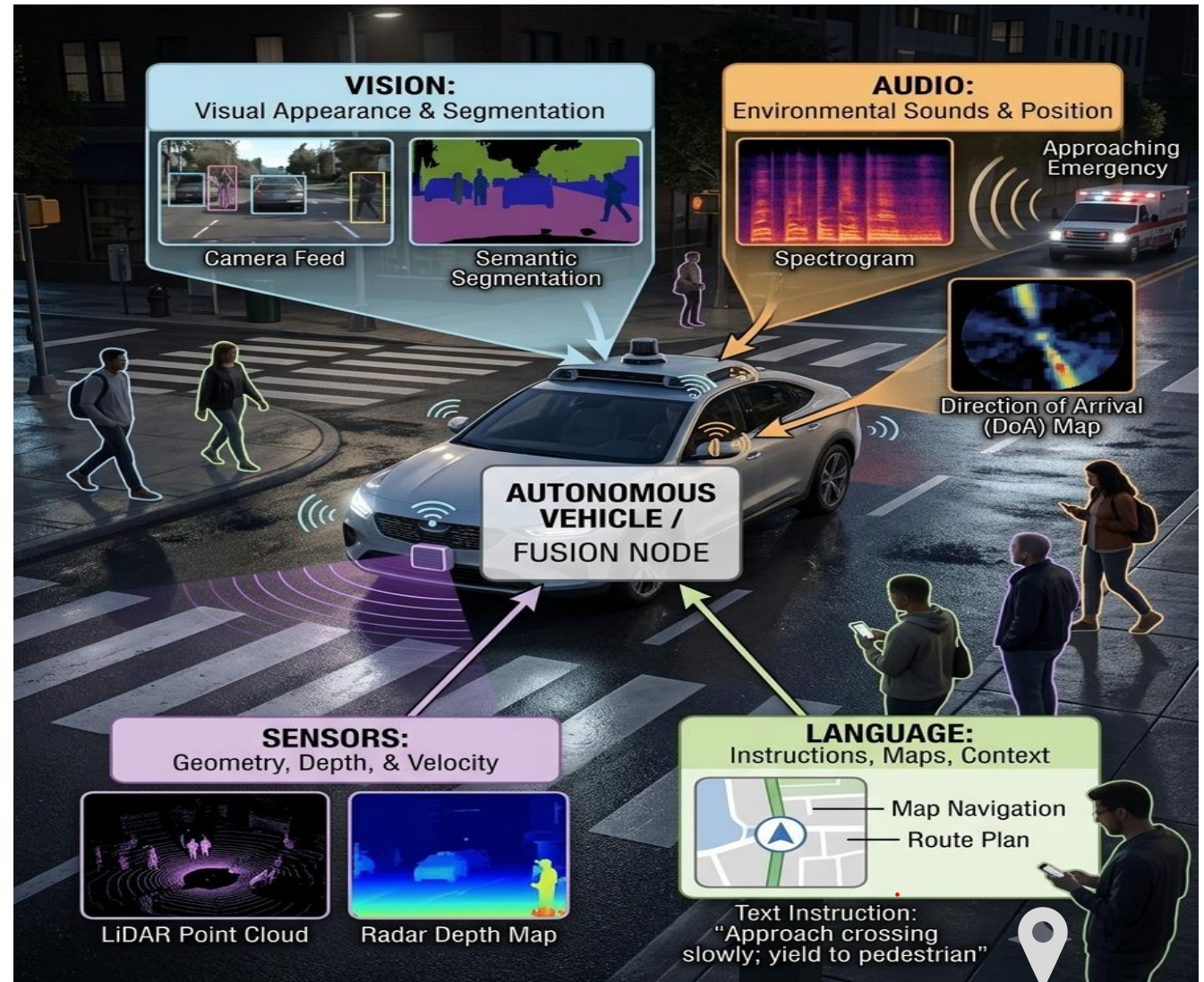
The challenge

Can one information source provide enough evidence to understand the situation reliably?

Real-world intelligence requires combining information that is distributed across different signals.

Think-pair-share

Which signal would you trust most in this situation?
When could that signal fail?



9 (ii): Why Multimodal AI?

The Central Question

How does intelligence emerge when information comes from multiple sources?

Traditional AI systems often specialize in one information type:

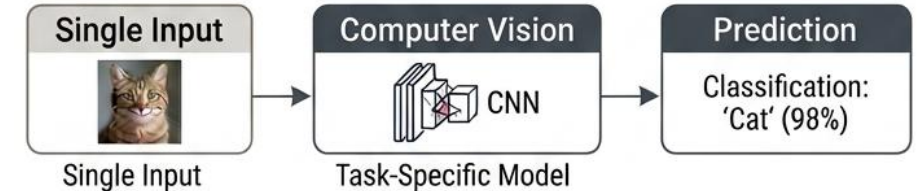
- **Computer Vision**
- **Natural Language Processing**
- **Speech AI**
- **Video Understanding**
- **Sensor AI**

But real-world intelligence rarely operates on one source alone.

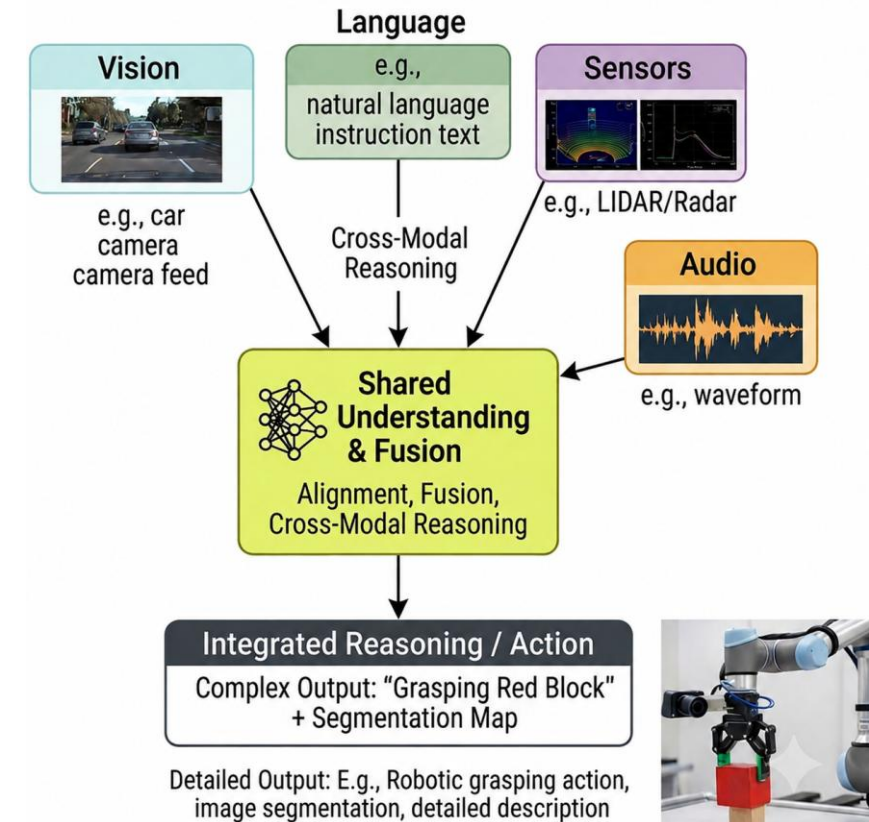
Key idea

The real world is multimodal. Intelligent systems therefore need mechanisms for integrating heterogeneous information.

SINGLE-MODAL AI



MULTI-MODAL AI

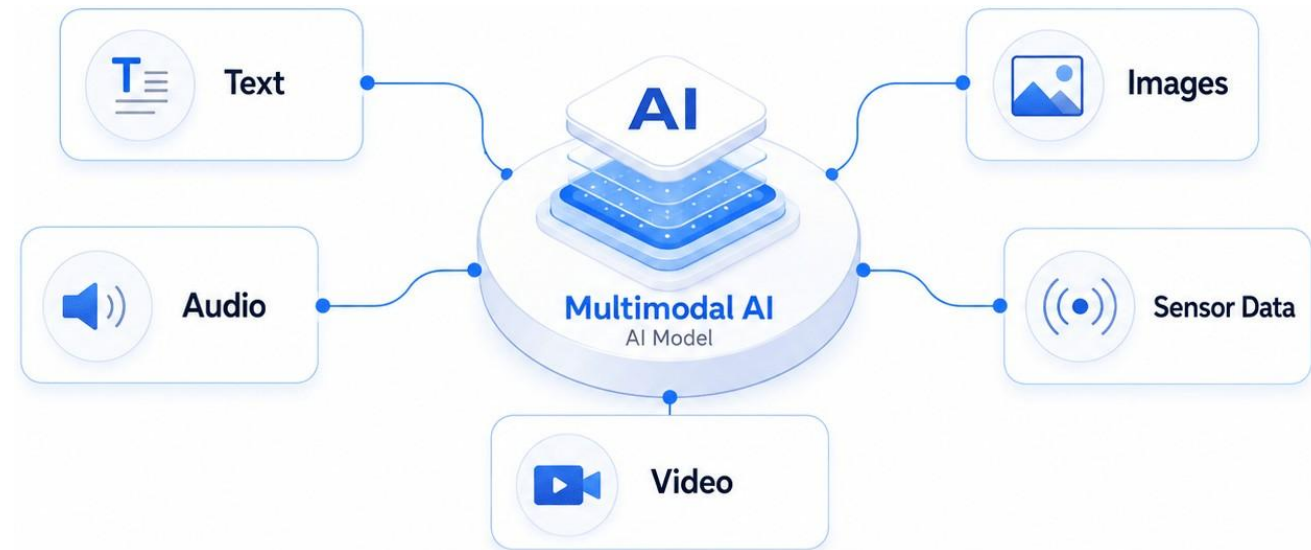
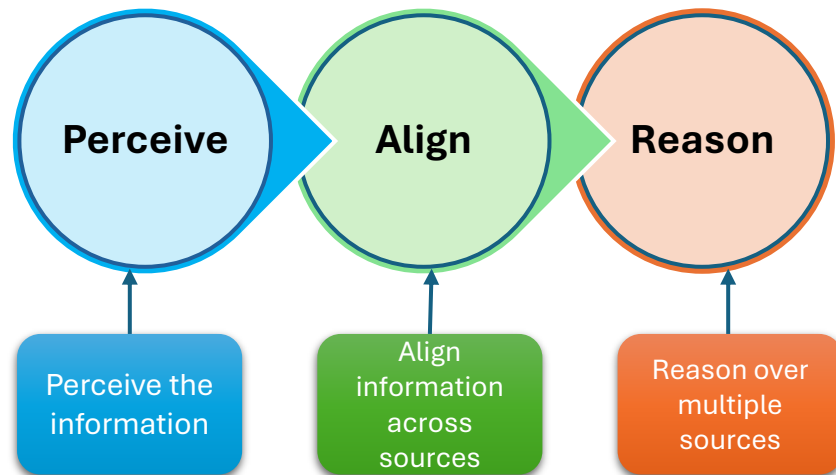


9 (iii): What is Multimodal AI?

Definition:

Multimodal AI refers to AI systems that can process, represent, connect, and reason over information originating from multiple modalities.

The course plan describes multimodal AI as systems that:



Important distinction

Multimodal AI is not simply putting several AI models together.

The central challenge is:

How can heterogeneous information be represented in a form that allows meaningful interaction between modalities?

9 (iv): Understanding "Modality"

Modality: A distinct channel or form through which information is represented or observed.

Common AI modalities:

Modality	Typical Data	Example
Vision	Images, pixels	 Photograph
Text	Words, documents	 Question
Audio	Waveforms, spectrograms	 Speech 
Video	Frames + time	 Human Activity 
Sensors	Measurements/signals	 Temperature, IMU ()
3D / Spatial	Point clouds, depth	 LiDAR 
Structured Data	Tables/events	 Patient Records 

Key insight

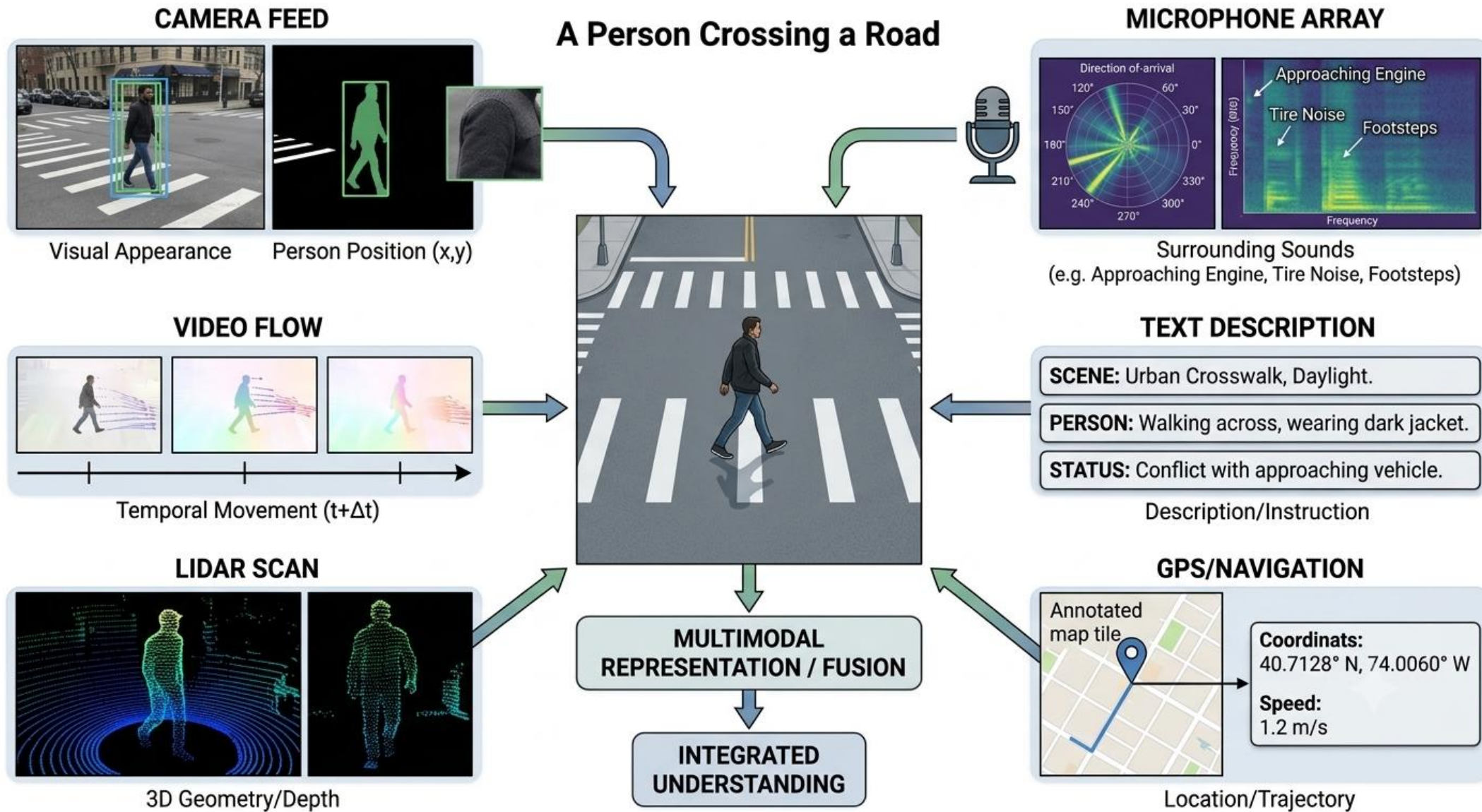
Different modalities describe the same world from different information perspectives.

For example:

A person crossing a road

The example is explained on the next slide with visualization.

9 (iv): Understanding "Modality" (Example)



9 (v): Why Are Modalities Different?

Different modalities have fundamentally different properties.

Vision

- Spatial structure
- High-dimensional
- Strong local relationships
- Rich visual semantics



Text

- Sequential / symbolic
- Highly semantic
- Discrete tokens



Audio

- Temporal signal
- Frequency structure
- Prosody and acoustic information (Tone, rhythm, and sound properties)



Video

- Spatial + temporal structure
- Motion
- Long-range temporal dependencies



Sensors

- Continuous/Numerical measurements
- Sampling rates
- Noise and missing values
- Physical-world constraints

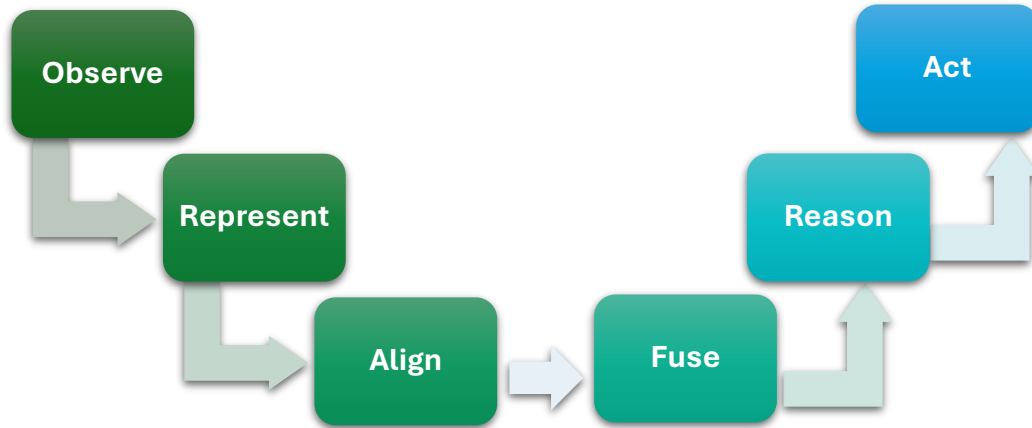


The challenge

A model cannot automatically assume that information represented differently is directly comparable. This is why multimodal systems require **representation learning, alignment, and fusion mechanisms**.

9 (vi): The Multimodal Pipeline

A useful mental model does the following:



Example:

Image + question

Image: a dog sitting beside a bicycle

Question:

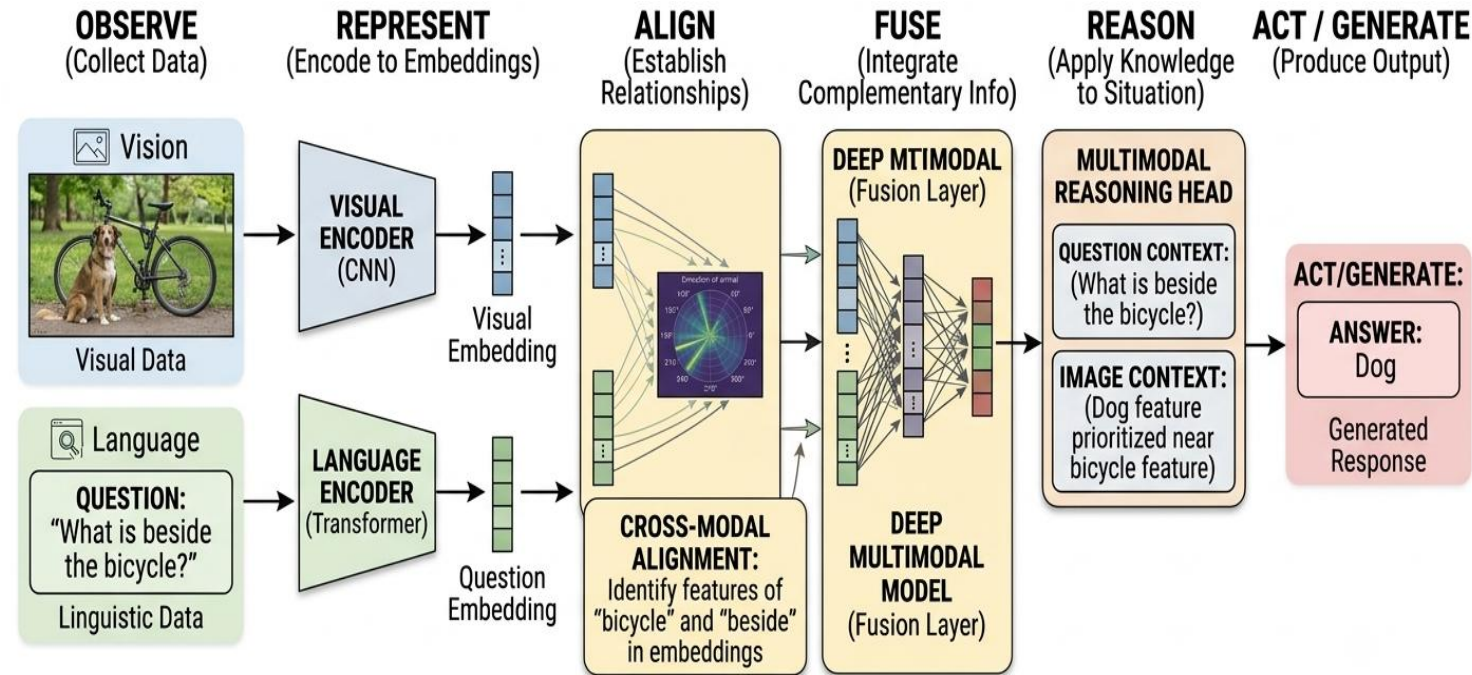
"What is beside the bicycle?"

System must:

Image → visual representation

Question → language representation → **Alignment** → **Multimodal reasoning**

Answer: **Dog**



9 (vii): The Three Core Concepts

Concept 1: Representation

- How do we represent multimodal data?

Before different modalities can interact, each must be transformed into a **shared computational form**.

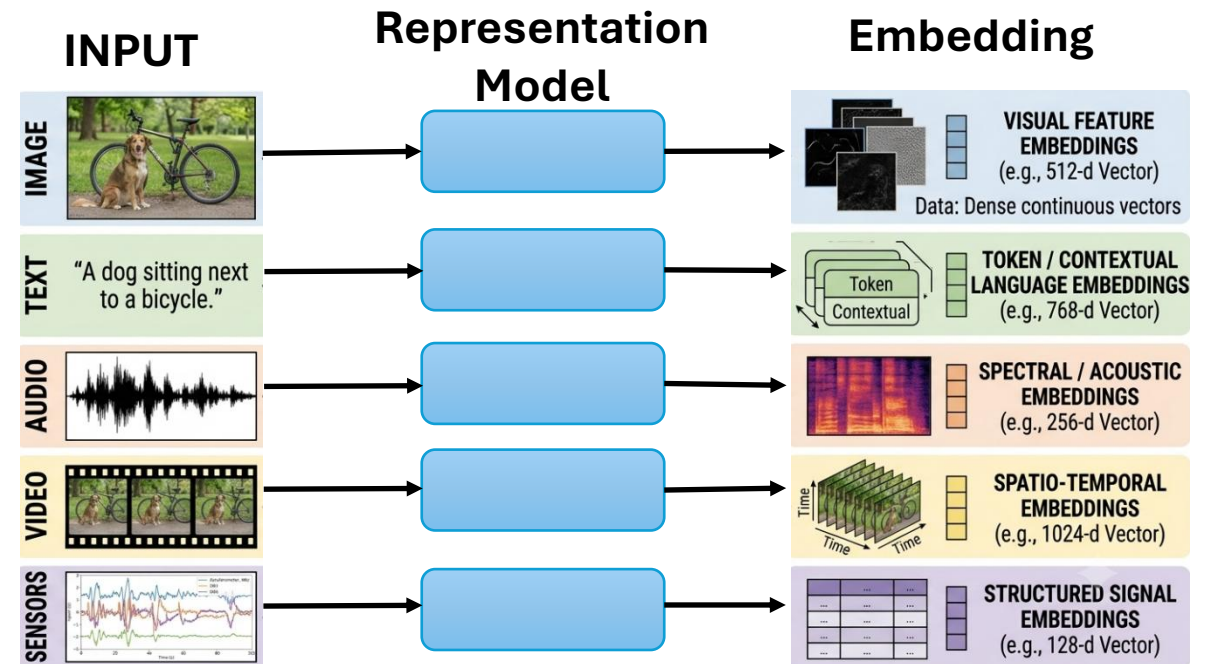
Raw inputs are encoded into **learned representations (embeddings)**:

- Why representation matters?

Representation is the **foundation layer** of multimodal learning:

Without meaningful representations, alignment and fusion are not possible.

It determines how well information from different modalities can later be compared and integrated.



9 (vii): The Three Core Concepts

What is an Embedding?

An **embedding** is a learned, dense vector representation that maps raw data into a continuous geometric space where semantic relationships are preserved.

Formally:

Semantically similar inputs → nearby vectors

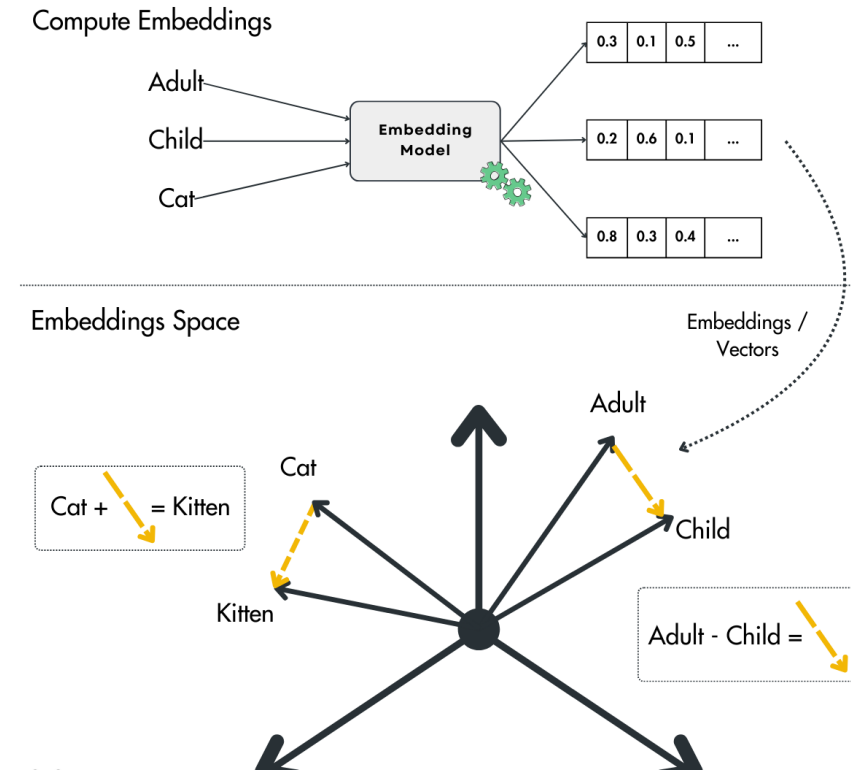
Semantically different inputs → distant vectors

This allows different types of data to be compared and processed in a unified mathematical space.

Example intuition:

“cat” (text) → vector
image of a cat → vector
sound of a cat → vector

- ❖ Although these inputs come from different modalities, their embeddings can still be placed close together in the model if they describe the same idea or concept.
- ❖ When the model learns well, embeddings form an organized “meaning space,” where similar concepts are located near each other, even if they come from different types of data (like text, images, or audio).



9 (vii): The Three Core Concepts

Concept 2: Alignment

What is multimodal alignment?

Multimodal alignment is the process of learning correspondences between elements across different modalities.

It answers:

➡ What in one modality refers to what in another?

Core idea

Alignment establishes cross-modal correspondence at the semantic or temporal level.

Why is alignment challenging?

Alignment is non-trivial because modalities differ in:

- Representation structure (continuous vs discrete)
- Temporal resolution (frame rate vs tokens vs signals)
- Spatial structure (pixels vs sequences vs vectors)
- Noise and missing information
- Level of abstraction (raw signal vs semantic meaning)

Example:

Video ↔ Action

Temporal frames correspond to events (e.g., *opening a door*)

Key distinction:

Representation defines *what each modality is*.

“Alignment defines *what corresponds across modalities*.”

9 (vii): The Three Core Concepts

Concept 3: Fusion

What is multimodal fusion?

Fusion is the process of integrating information from multiple modalities into a unified representation or decision space.

It answers:

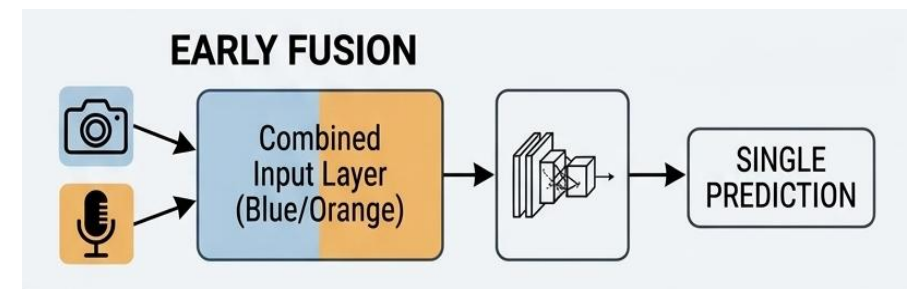
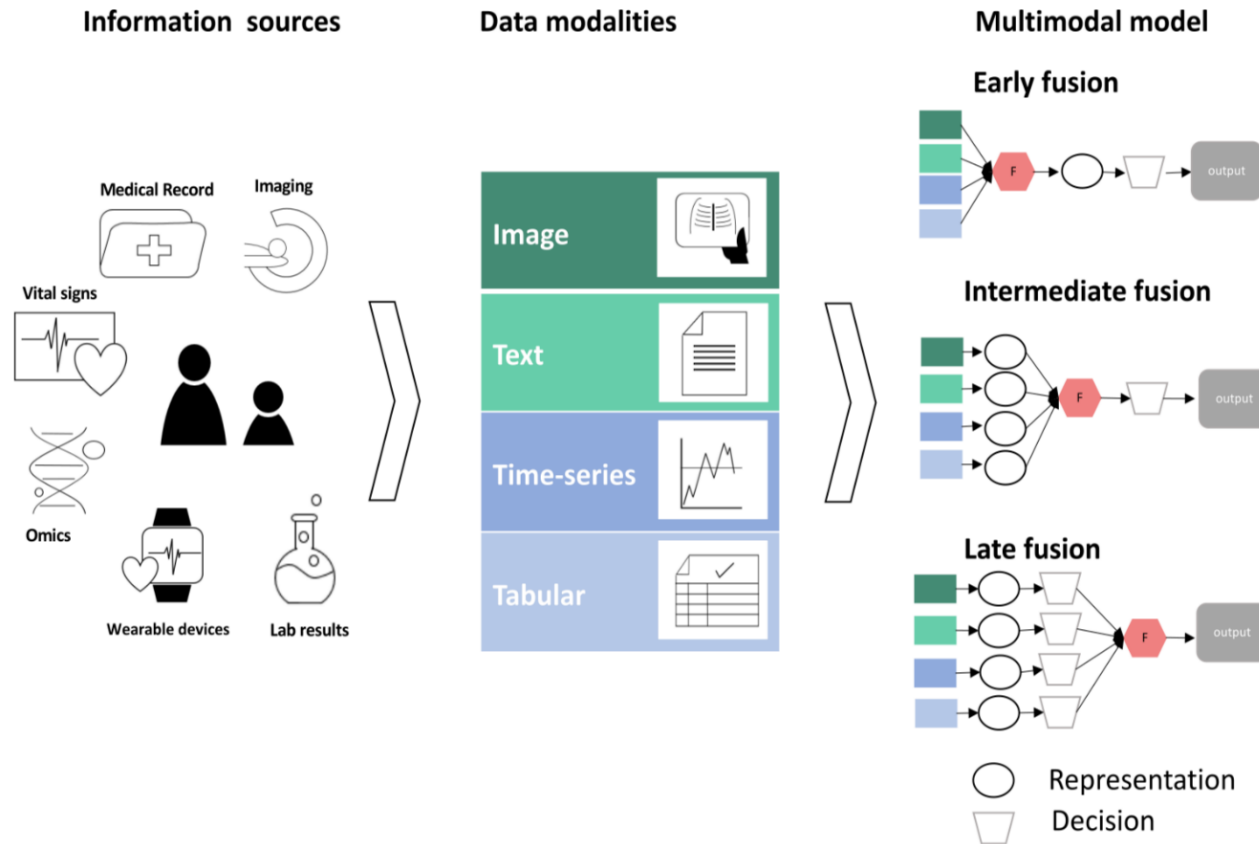
How should aligned information be combined for reasoning or prediction?

Types of fusion

1. Early Fusion (Input-level integration)

Raw or low-level features are combined before deep processing.

- Image features + text features → joint input representation
 - ✓ Simple
 - ✗ Limited modality-specific processing

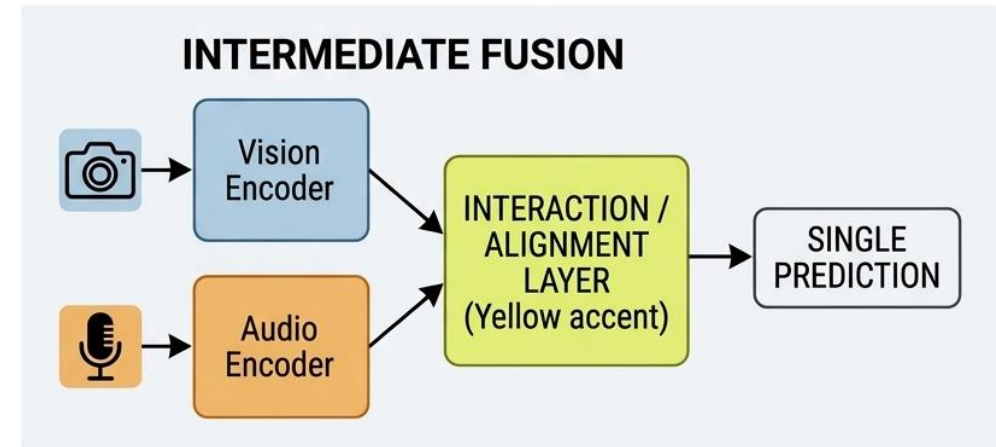


9 (vii): The Three Core Concepts

2. Intermediate Fusion (Representation-level interaction)

Each modality is encoded separately, then combined through interaction layers.

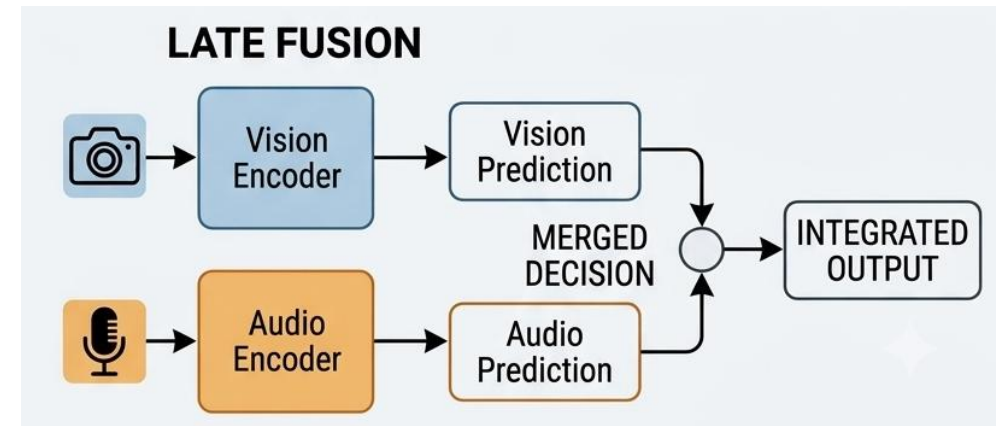
- Vision encoder + Language encoder → cross-modal attention / interaction
 - ✓ Most widely used in modern multimodal models
 - ✓ Preserves modality-specific structure



3. Late Fusion (Decision-level integration)

Each modality produces independent predictions, which are then combined.

- Vision prediction + Audio prediction → final decision
 - ✓ Robust to missing modalities
 - ✗ Limited deep interaction



Key principle

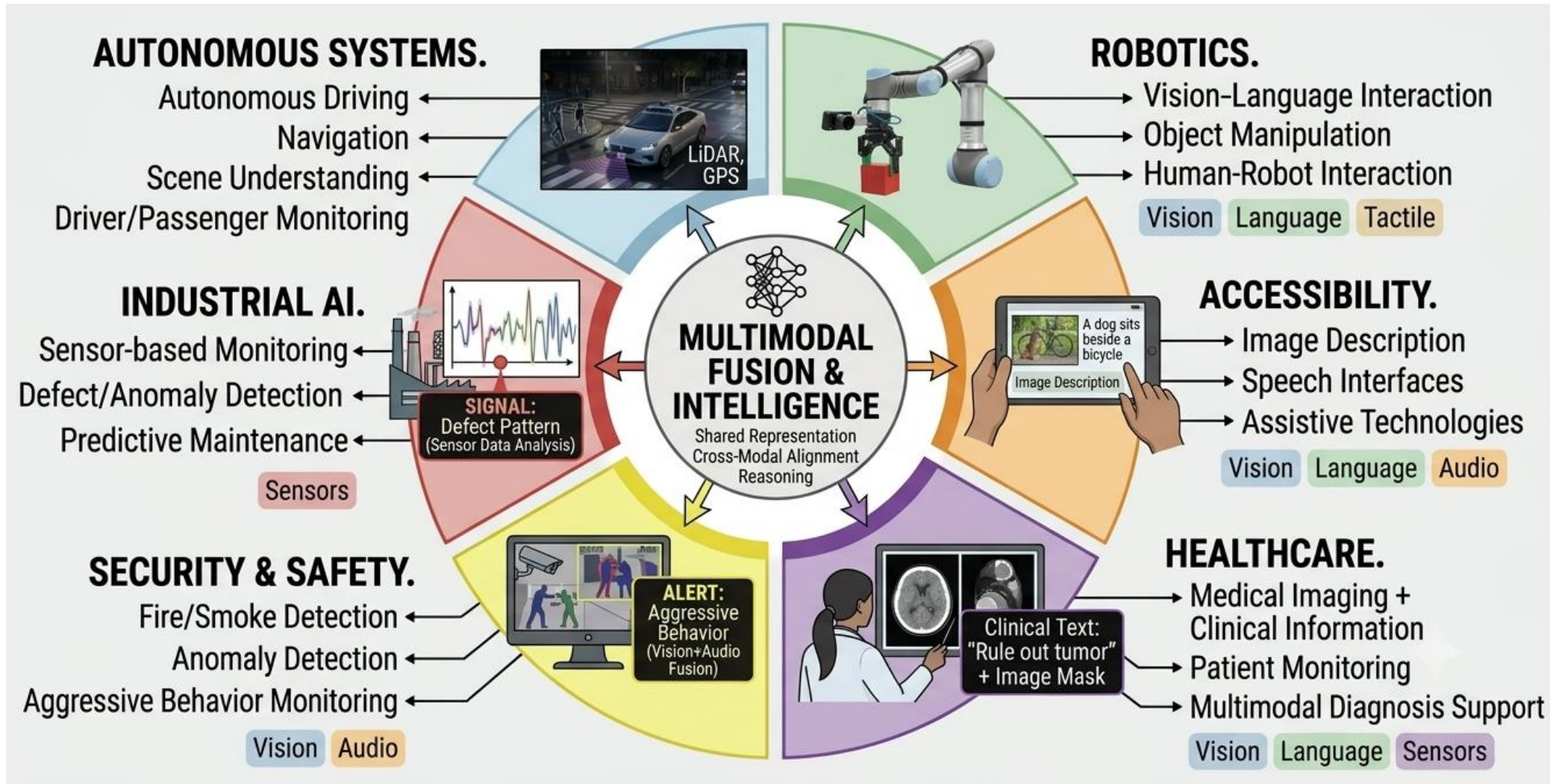
Fusion is most effective when modalities provide:

- **Complementary information** (different aspects of same event)
- **Redundant information** (confirmation across sources)
- **Contextual information** (one modality *disambiguates* another)

9 (viii): Application Landscape

Where Is Multimodal AI Used?

Multimodal AI spans multiple real-world domains.



9 (ix): Course Case Studies

From Theory to Real Systems

The course is structured to bridge foundational multimodal AI theory with real-world intelligent systems deployed in perception, monitoring, and decision-making tasks.

Each case study represents a progression in system complexity — from low-level perception to high-level reasoning and action.

❑ LOW-LEVEL PERCEPTION SYSTEMS:

1. Image Enhancement & Restoration

Focus:

- Improving visual quality and signal fidelity
- Noise removal, deblurring, and reconstruction
- Foundational step for all vision-based AI systems



Fig. 2: (a) Sample images of LoLI-Street. **Green:** train and validation sets, **Red:** real low-light test set. (b) Distribution of the low-light images of our LoLI-Street.

[LoLI-Street: Benchmarking Low-Light Image Enhancement and Beyond](#), The 17th Asian Conference on Computer Vision (ACCV), 2024, Hanoi, Vietnam. BK IF=1.

9 (ix): Course Case Studies

From Theory to Real Systems

❑ LOW-LEVEL PERCEPTION SYSTEMS:

1. Image Enhancement & Restoration
2. Haze-Aware Image Dehazing

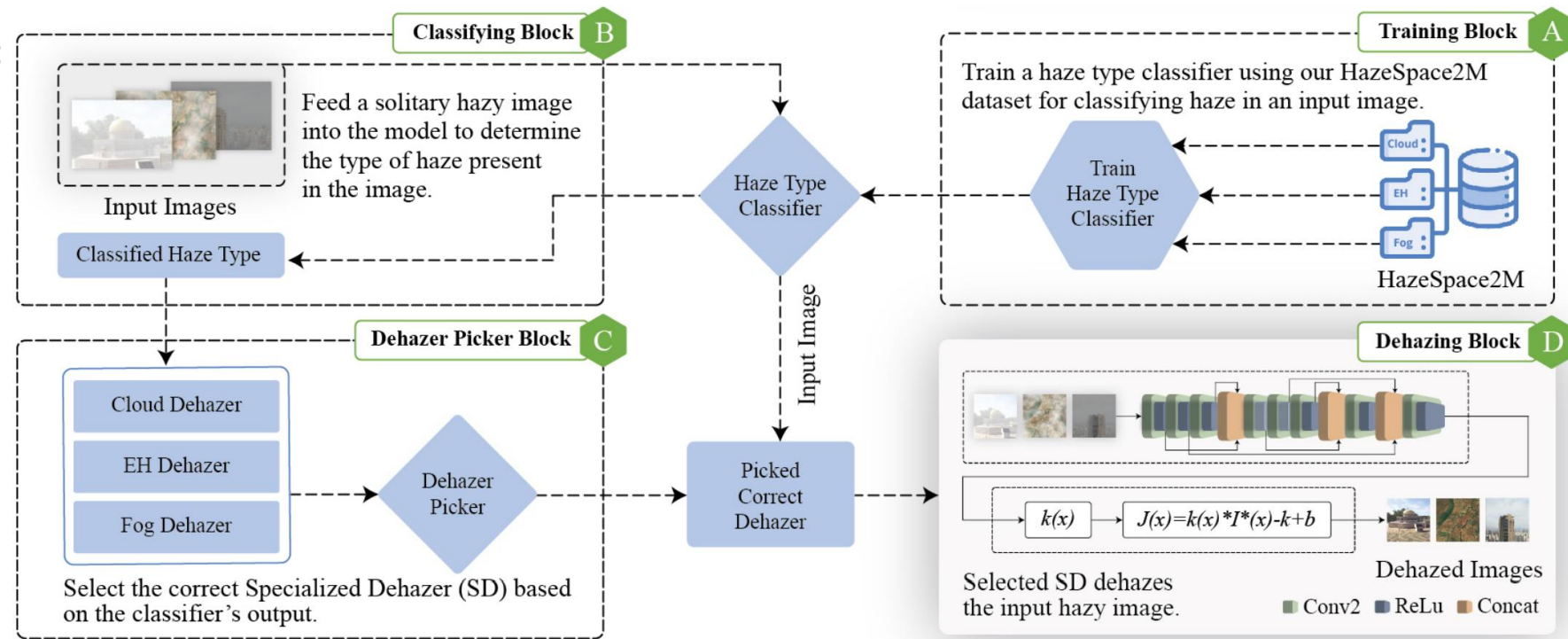


Fig. 2: Our proposed framework for haze-aware single-image dehazing using specialized dehazers.

Dataset: HazeSpace2M

- ❖ A large-scale paired dehazing dataset
- ❖ Features over 2 million hazy images

Key Features

- ❖ Features four different scene types
- ❖ Includes three different haze types

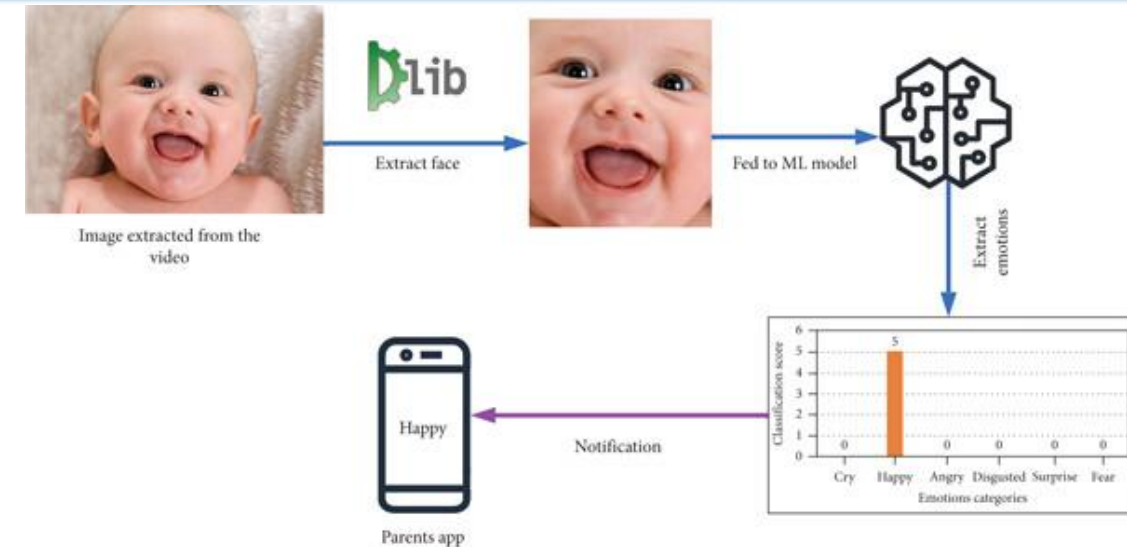
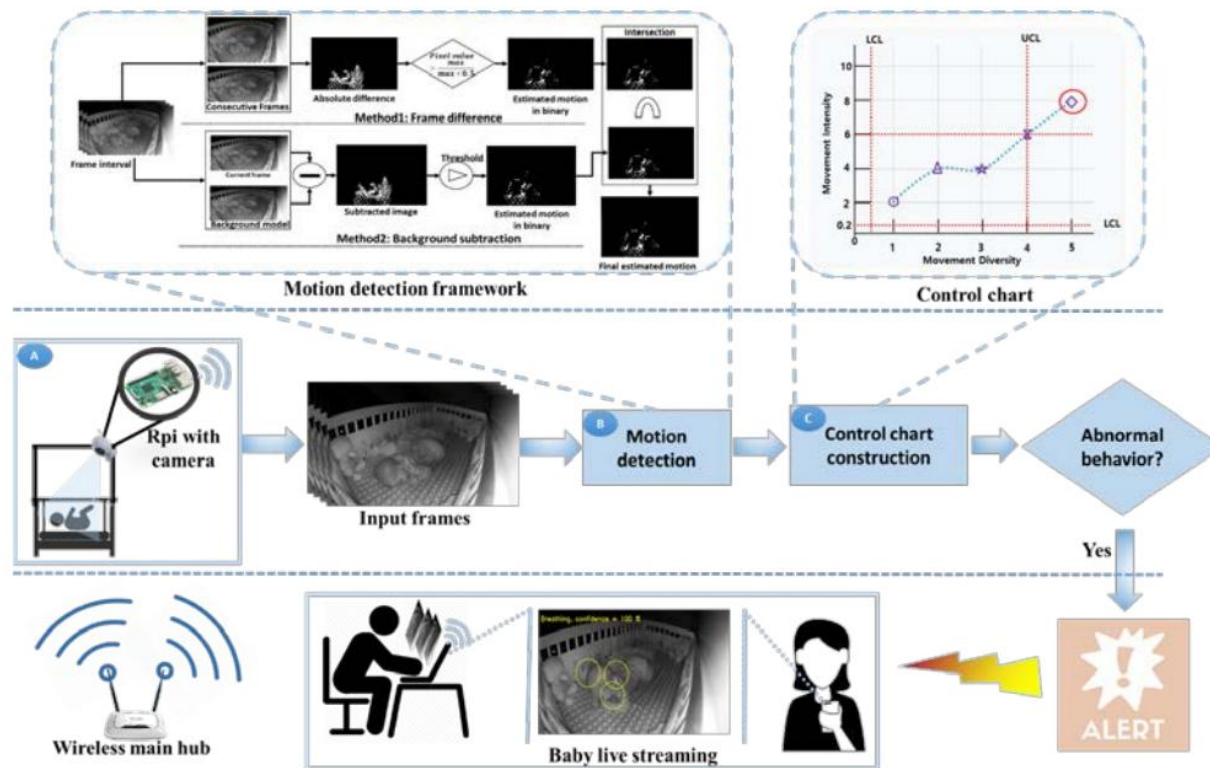
Course Case Studies: From Theory to Real Systems

ENVIRONMENTAL UNDERSTANDING SYSTEMS

1. Intelligent Baby Monitoring

Focus:

- Continuous multimodal scene understanding
- Detection of activity patterns and anomalies
- Early-stage event recognition in dynamic environments



Proposed baby behaviour monitoring framework. (A): Input is acquired from smart camera attached to RPi, (B): motion computation where two different algorithms are executed to compute motion of baby, and (C): control chart construction from the detected motion, abnormal breathing behaviour detection based on one interval of control chart, and alert transmission in case of any abnormal breathing or motion behaviour.

Hussain, T., **Muhammad, K.**, Khan, S., Ullah, A., Lee, M. Y., & Baik, S. W. (2019). [Intelligent baby behavior monitoring using embedded vision in IoT for smart healthcare centers](#). *Journal of Artificial Intelligence and Systems*, 1(1), 110-124.

Course Case Studies: From Theory to Real Systems

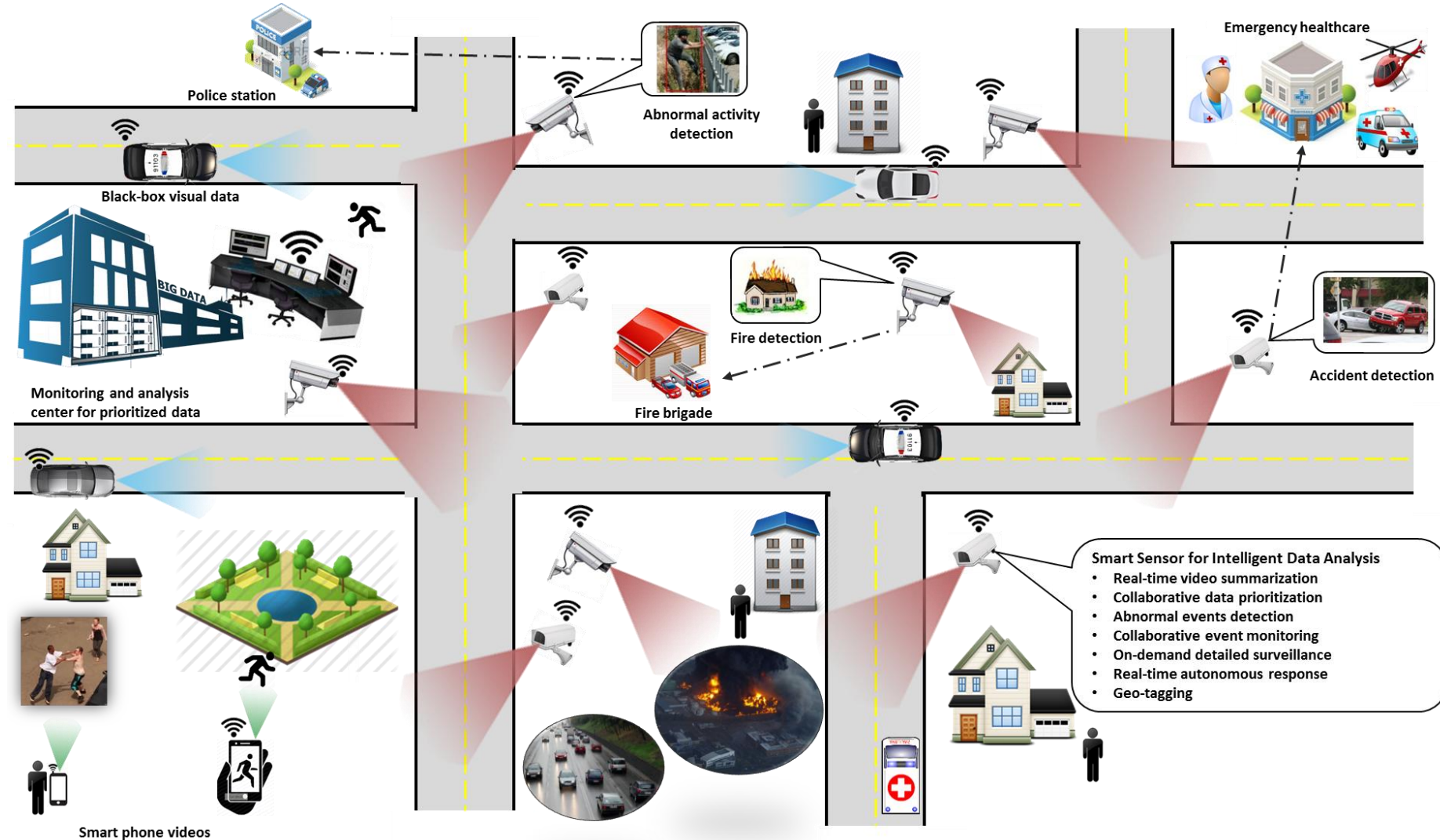
ENVIRONMENTAL UNDERSTANDING SYSTEMS

1. Intelligent Baby Monitoring

2. Fire & Smoke Detection

Focus:

- Real-time scene interpretation
- Safety-critical visual recognition
- Early warning through visual cues and environmental signals



Course Case Studies: From Theory to Real Systems

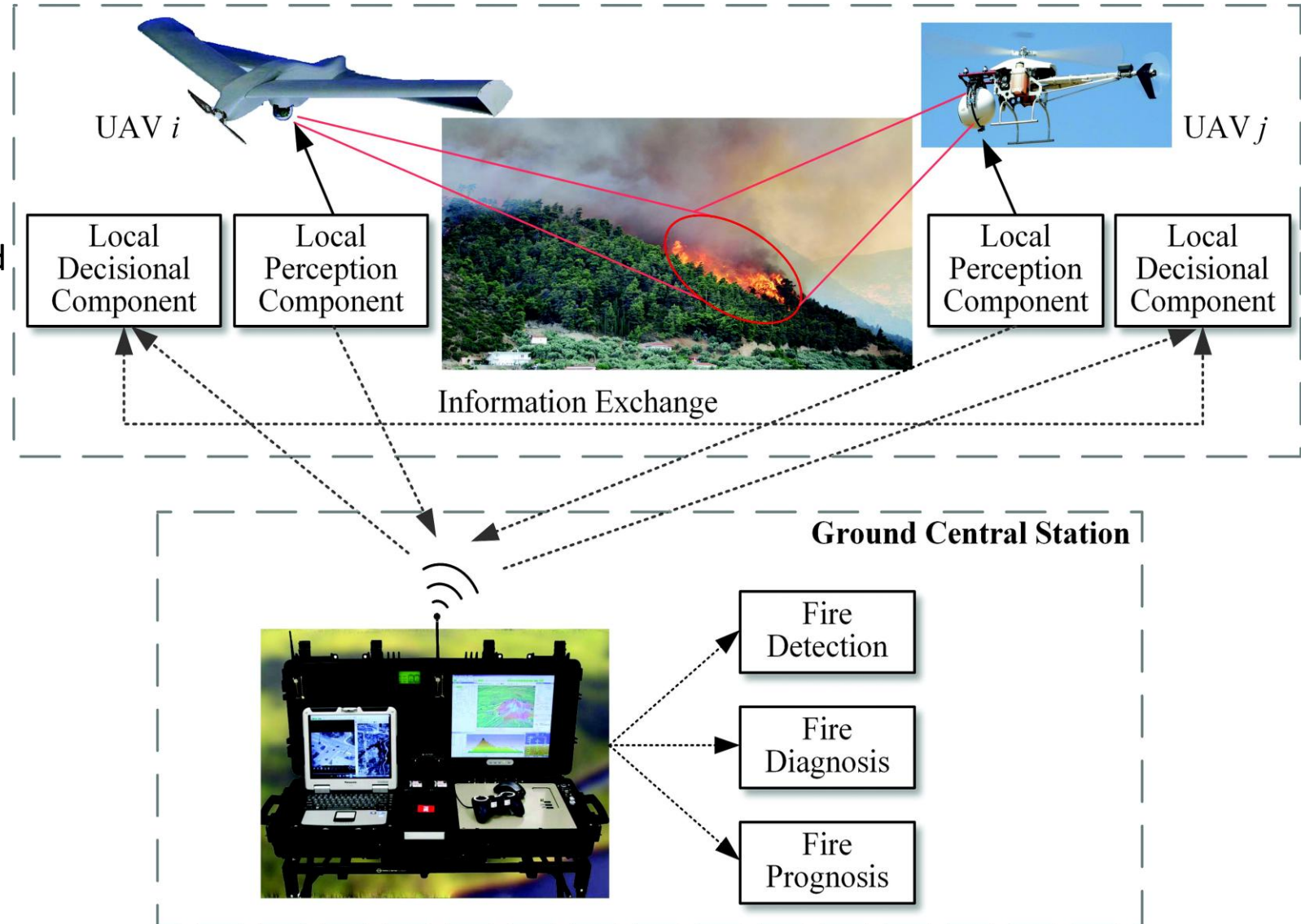
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Course Case Studies: From Theory to Real Systems

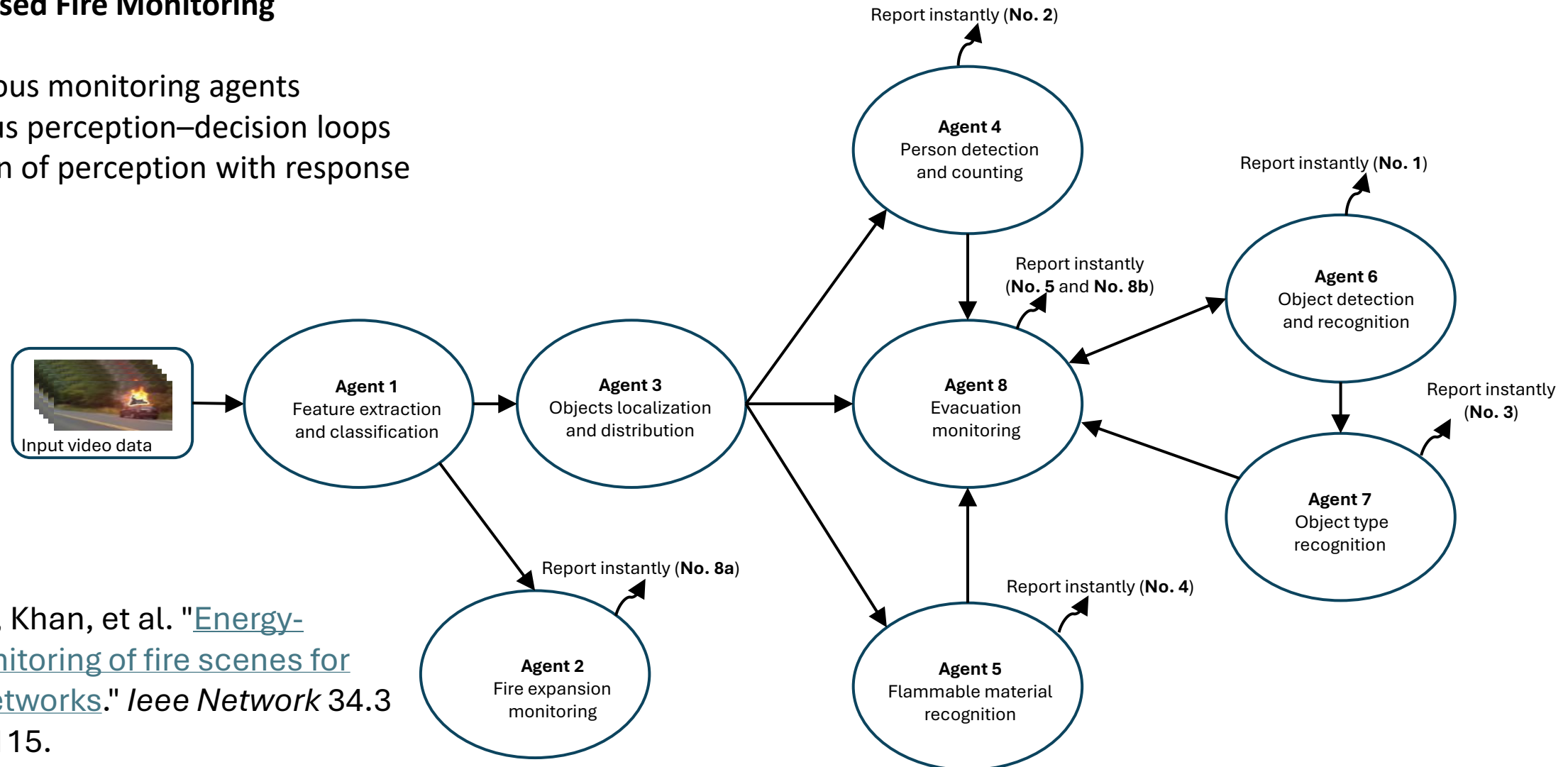
AGENTIC & DECISION-ORIENTED SYSTEMS

1. Agent-Based Fire Monitoring

Focus:

- Autonomous monitoring agents
- Continuous perception–decision loops
- Integration of perception with response strategies

Collaborative monitoring of fire scene through agents



Muhammad, Khan, et al. "[Energy-efficient monitoring of fire scenes for intelligent networks](#)." *Ieee Network* 34.3 (2020): 108-115.

Course Case Studies: From Theory to Real Systems

❑ BEHAVIOR & HUMAN ACTIVITY INTELLIGENCE

1. Human Action Recognition

Focus:

- Understanding structured human motion patterns
- Temporal modeling of behavior
- Multimodal sensor-based interpretation

CHILD: Child Hazardous Indoor Labeled Dataset for Activity Recognition

ACM Multimedia, 2025, BK IF=4.

<https://dl.acm.org/doi/abs/10.1145/3746027.3754795>



Figure 3: Example frames illustrating various activities from the CHILD dataset, categorized by risk level. Activities begin with normal activity, transition through low-risk and medium-risk, and conclude with high-risk.

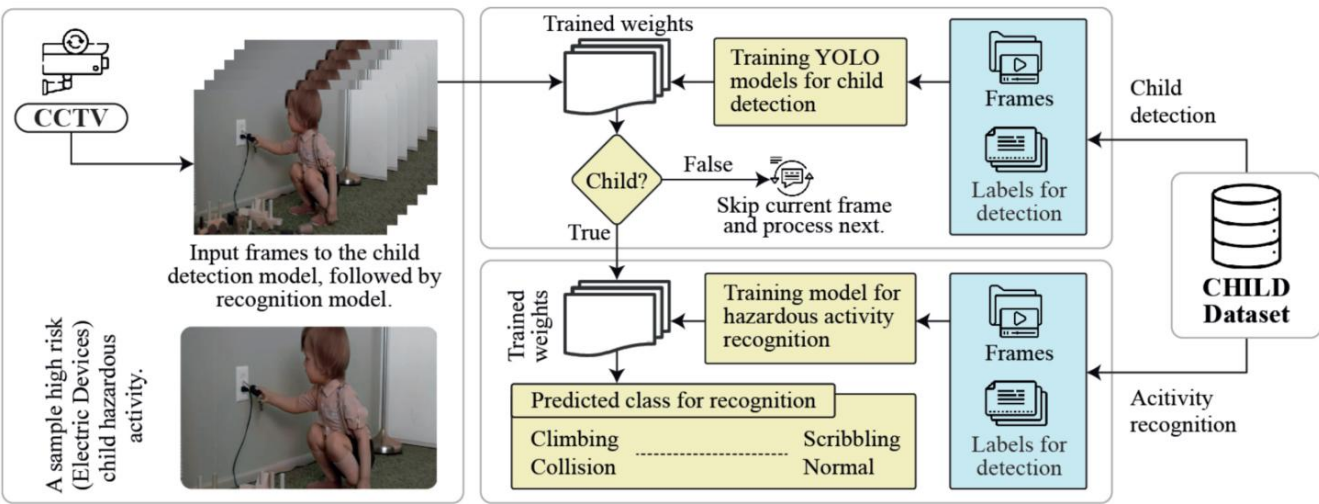
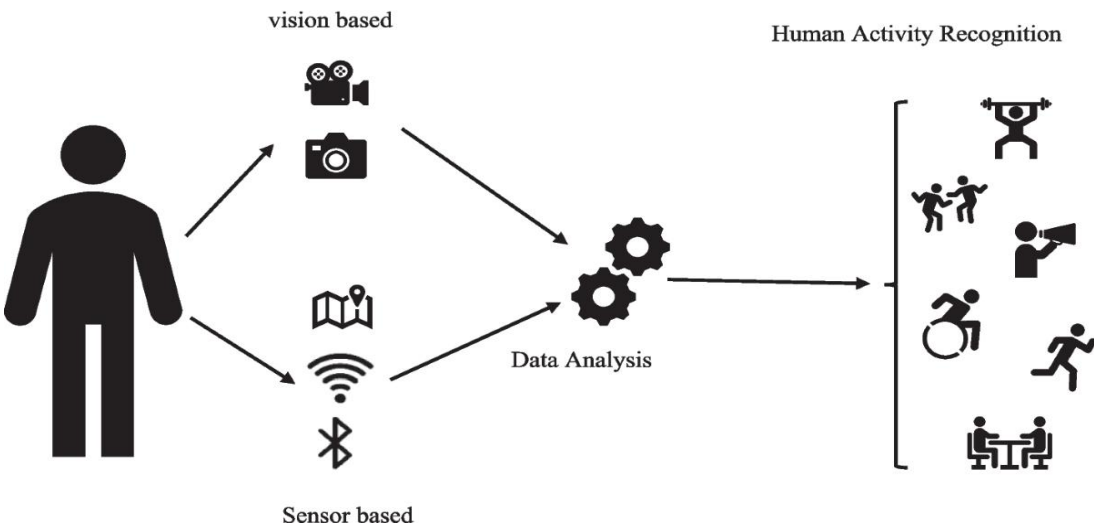


Figure 4: Proposed HazardNet: A custom framework for monitoring child-hazardous activities in indoor settings, incorporating dedicated detection and recognition models for effective performance, utilizing the newly introduced CHILD dataset.

Course Case Studies: From Theory to Real Systems

❑ BEHAVIOR & HUMAN ACTIVITY INTELLIGENCE

1. Human Action Recognition

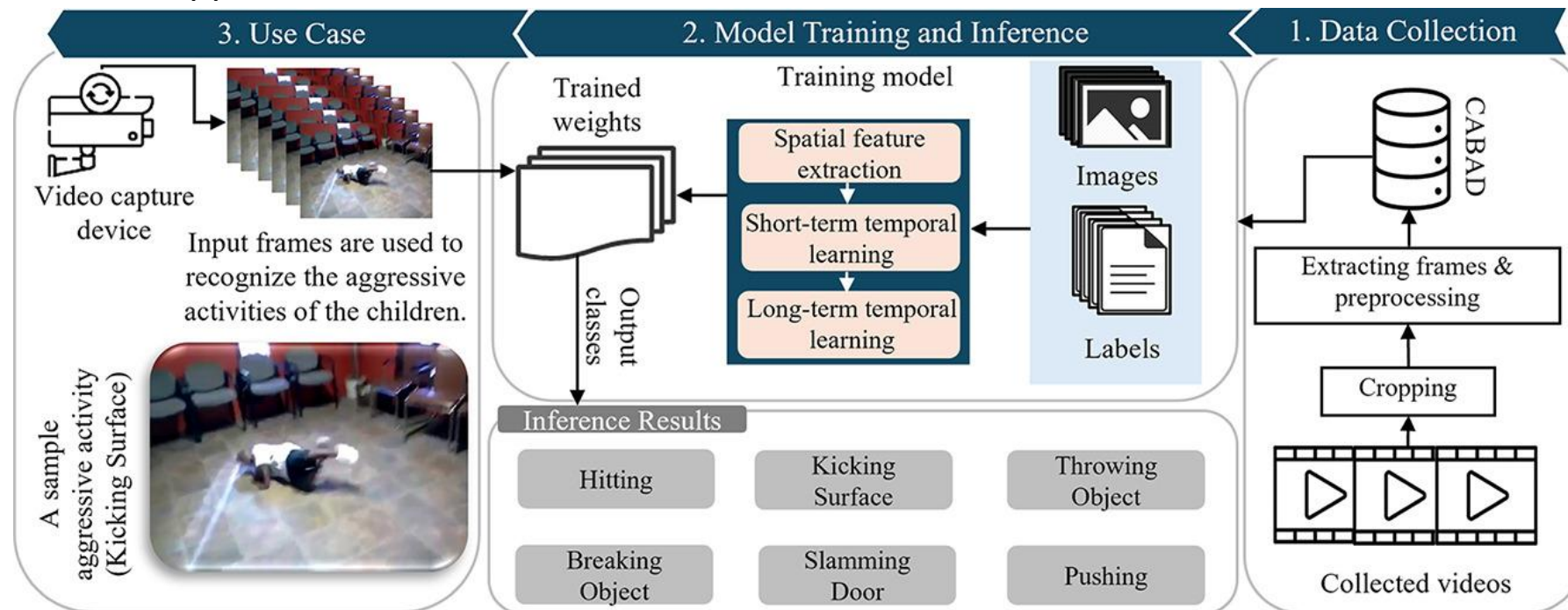
2. Aggressive Behavior Monitoring

Focus:

- Complex behavior classification in real environments
- Context-aware interpretation of human actions
- Safety and surveillance applications

[CABAD: A Video Dataset for Benchmarking Child Aggression Recognition](#)

Alexandria Engineering Journal (Impact Factor: 6.2, Q1 Rank, Top 5%)



Child Aggression Behavior Analysis Dataset (CABAD) and CABA_Net

Course Case Studies: From Theory to Real Systems

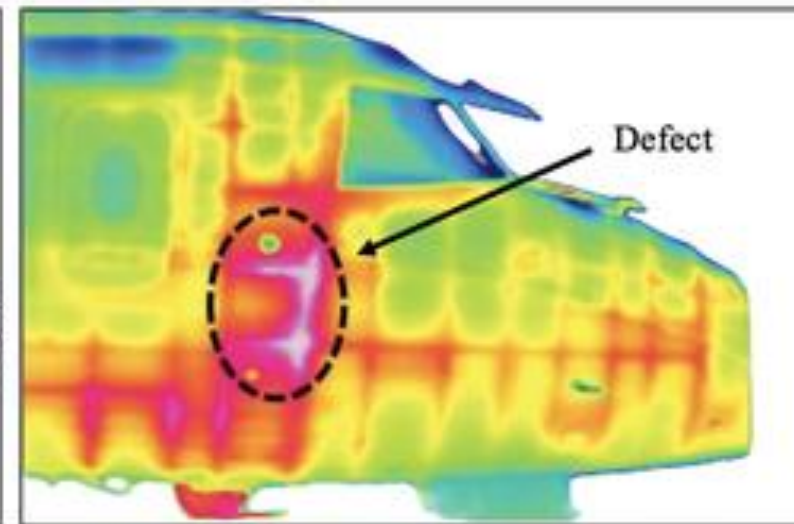
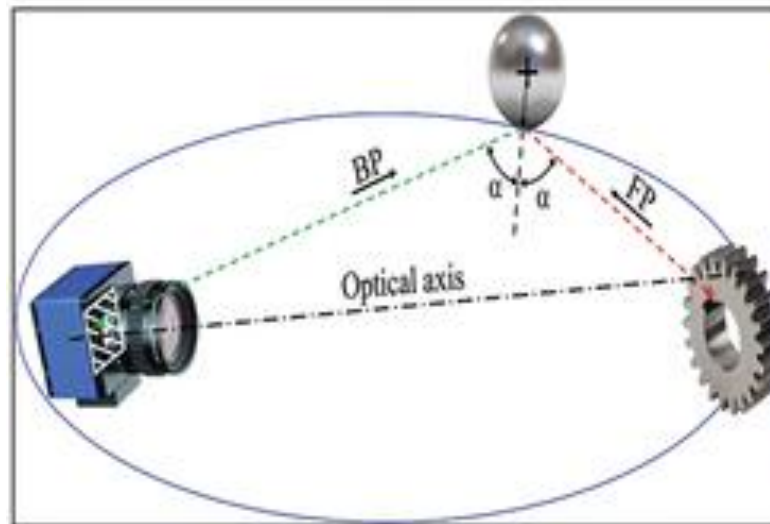
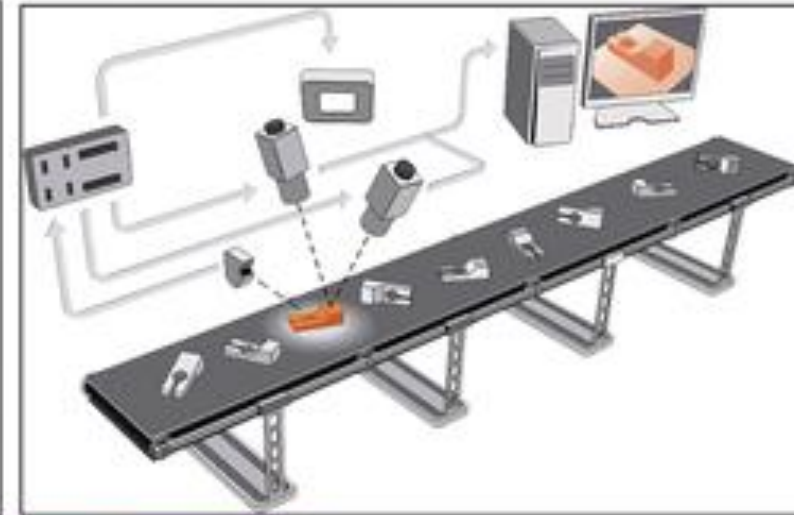
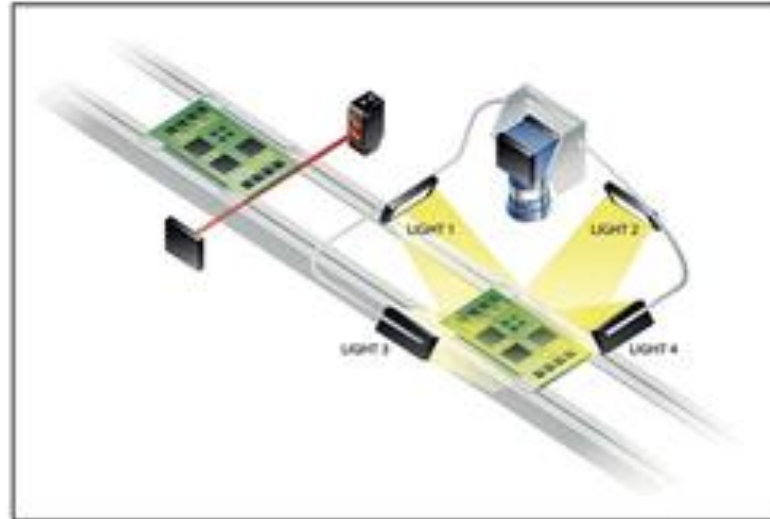
❑ BEHAVIOR & HUMAN ACTIVITY INTELLIGENCE

❑ INDUSTRIAL INTELLIGENCE SYSTEMS

1. Industrial Anomaly Detection

Focus:

- Detection of deviations in operational systems
- Multisensor pattern analysis
- Predictive monitoring for system reliability



Course Case Studies: From Theory to Real Systems

IARD: Intruder Activity Recognition Dataset for Threat Detection

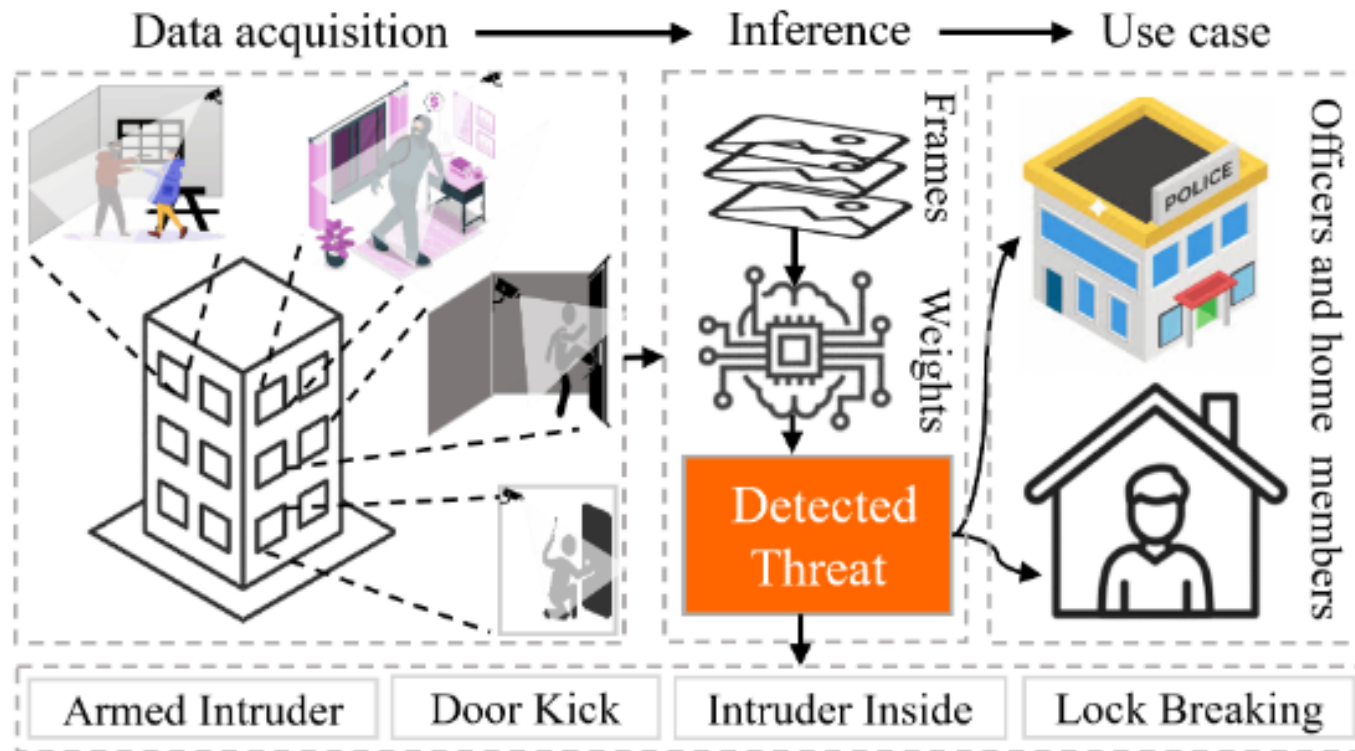


Figure 1: A High-level overview of a surveillance system integrating AI methods learned from our IARD dataset to recognize intruder threats captured by cameras.

The 34th ACM Conference on Information and Knowledge Management (CIKM), COEX, Seoul, 2025, BK IF=3.



Figure 2: Sample frames from the IARD dataset.

Course Case Studies: From Theory to Real Systems

ROAD-6: A Diverse Dataset for Unexpected Hazard Recognition in Autonomous Vehicles



The 15th ACM International Conference on Multimedia Retrieval (ICMR), USA. BK IF=1.

Figure 1: Sample ROAD-6 images depicting risk levels: High (red), Moderate (orange), and Low (blue).

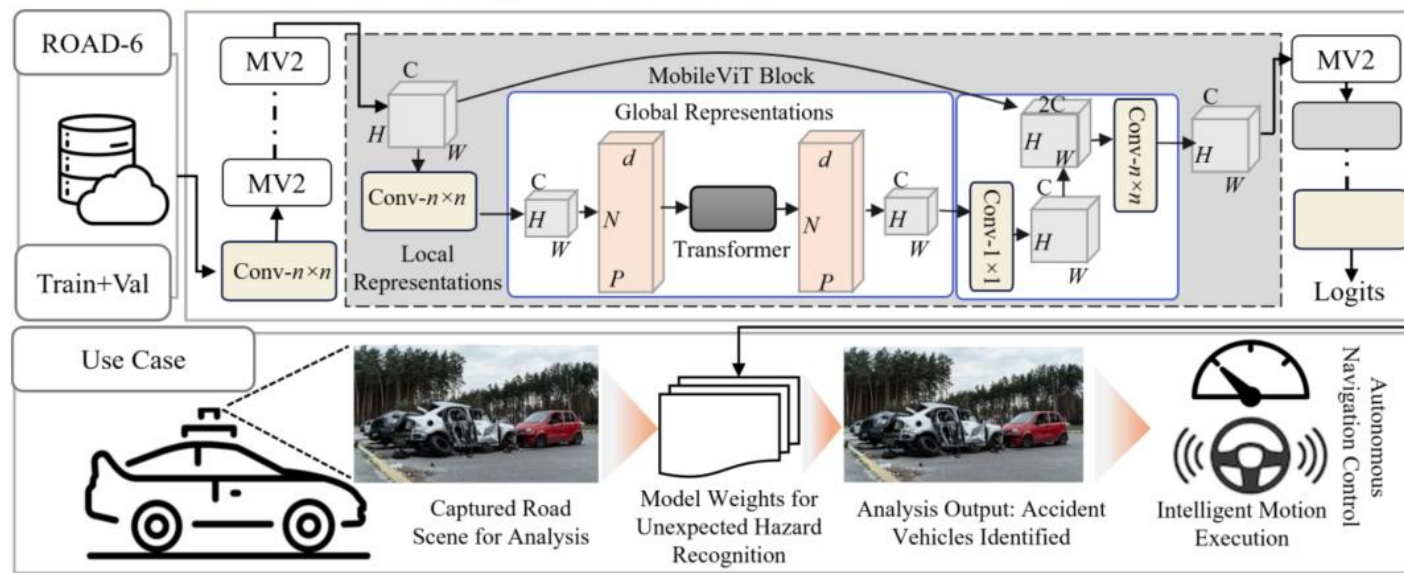


Figure 2: Use case of unexpected hazard recognition framework for AV utilizing MobileViT and the ROAD-6 dataset.

9 (x): Course Learning Outcomes

1. Explain

Explain the fundamental concepts behind multimodal AI.

3. Align

Explain how information from different modalities can be associated and aligned.

5. Analyze architectures

Understand the role of neural networks, embeddings, attention, and Transformers in multimodal systems.

7. Reading Literature

Understand and critically discuss multimodal AI research papers and case studies.

By the end of the course, students should be able to understand these

2. Represent

Understand how different modalities are represented computationally.

4. Fuse

Understand major strategies for combining multimodal information.

6. Interpret applications

Analyze how multimodal AI is used in vision, audio, video, sensors, robotics, healthcare, autonomous systems, and safety applications.

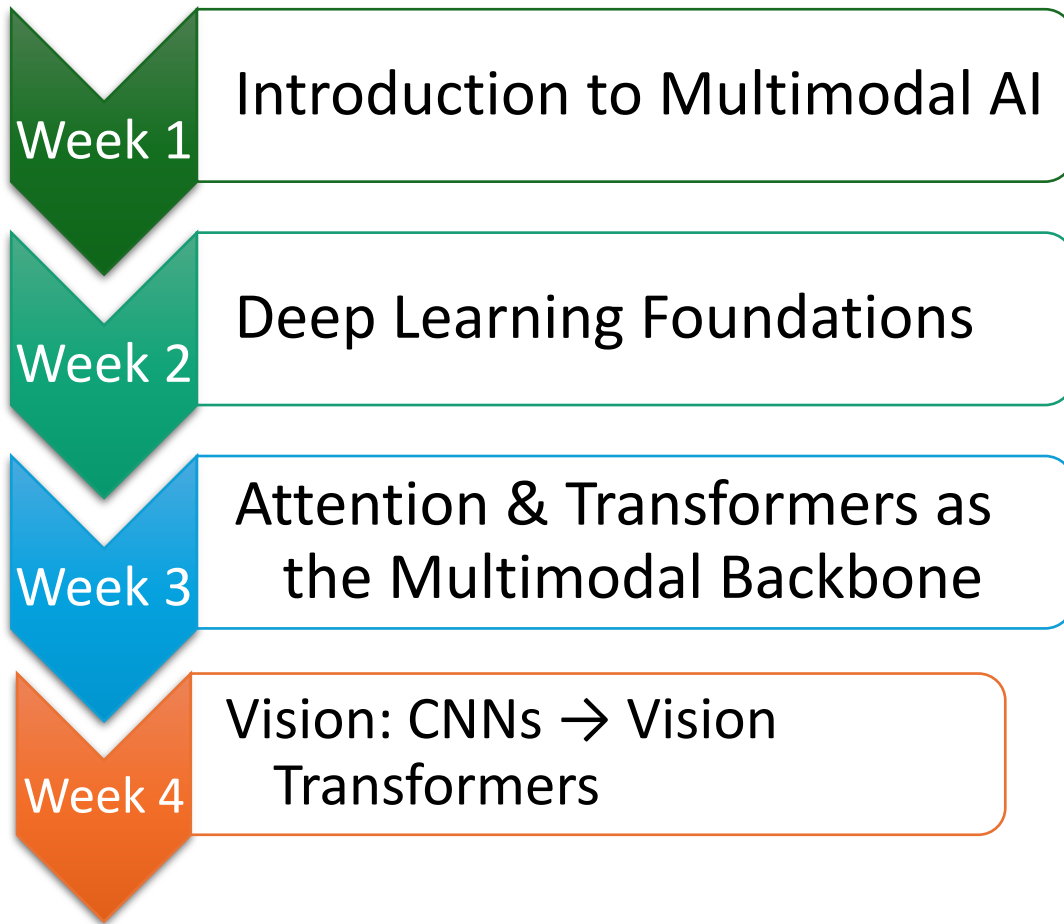
8. Identify limitations

Recognize challenges such as noise, missing modalities, alignment errors, computational cost, bias, and reliability.

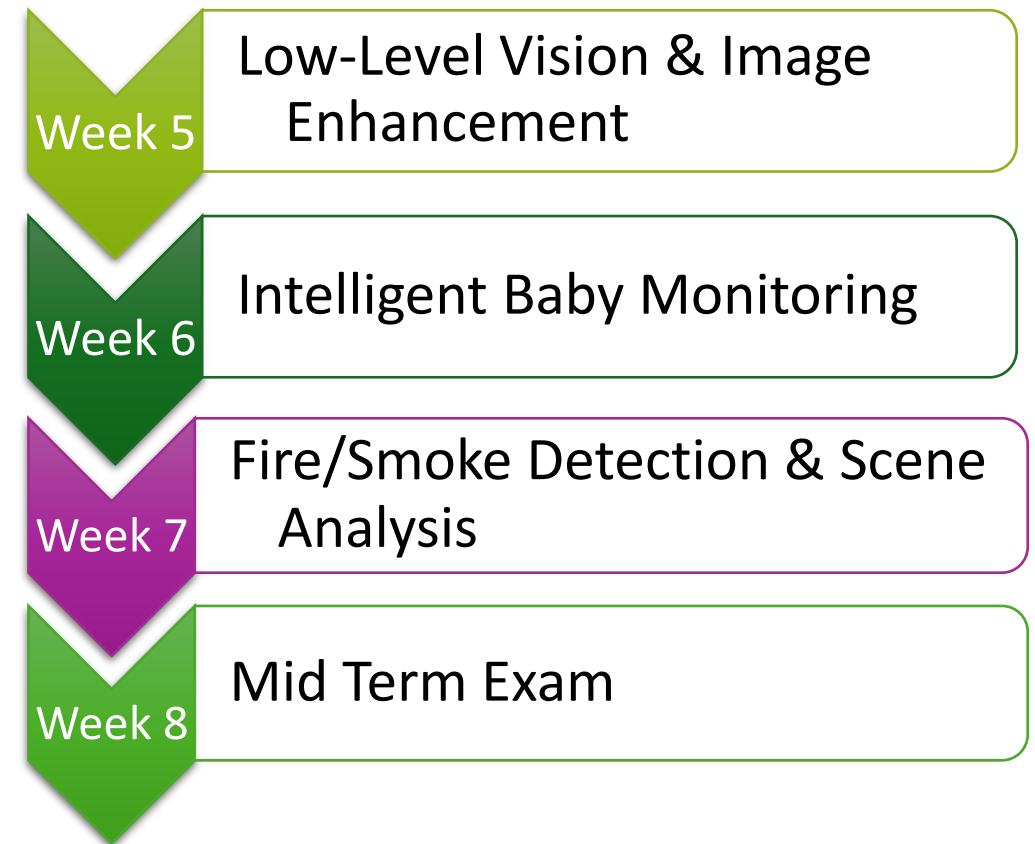


9 (xi): Course Roadmap

Phase: 1 Foundations

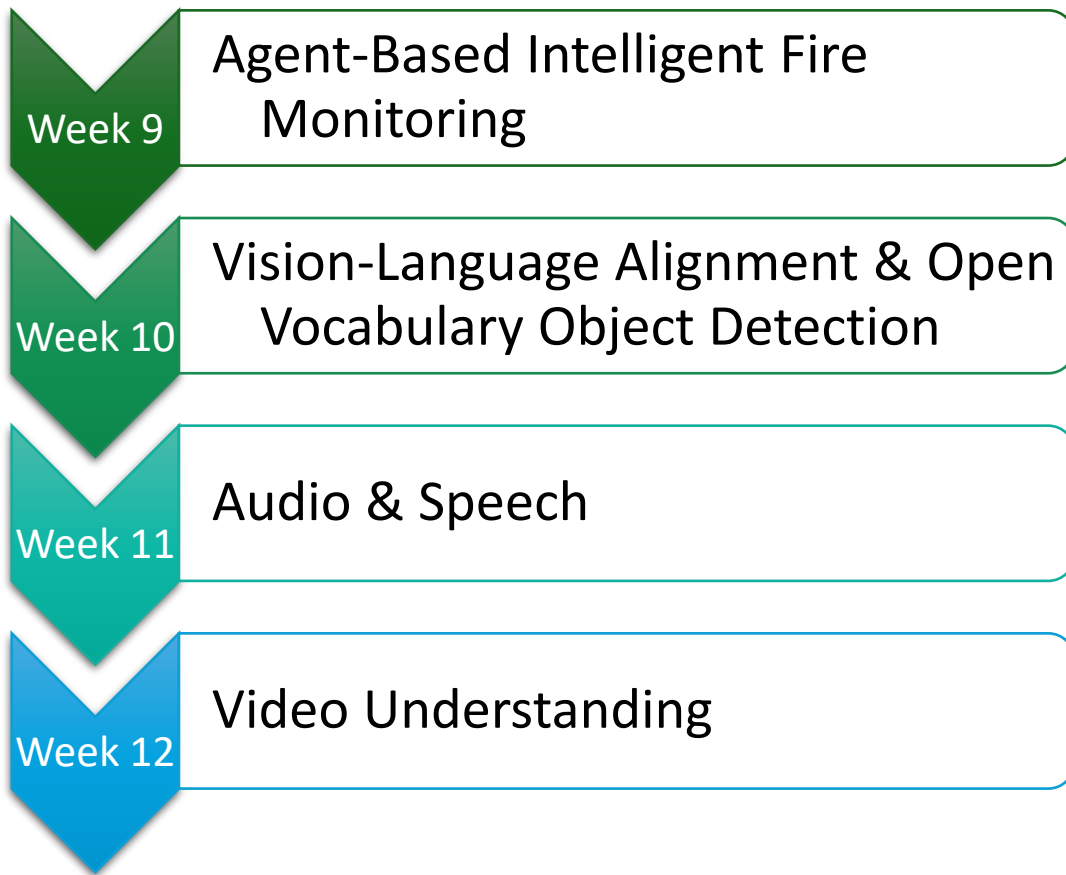


Phase 2 Vision & Real-World Perception

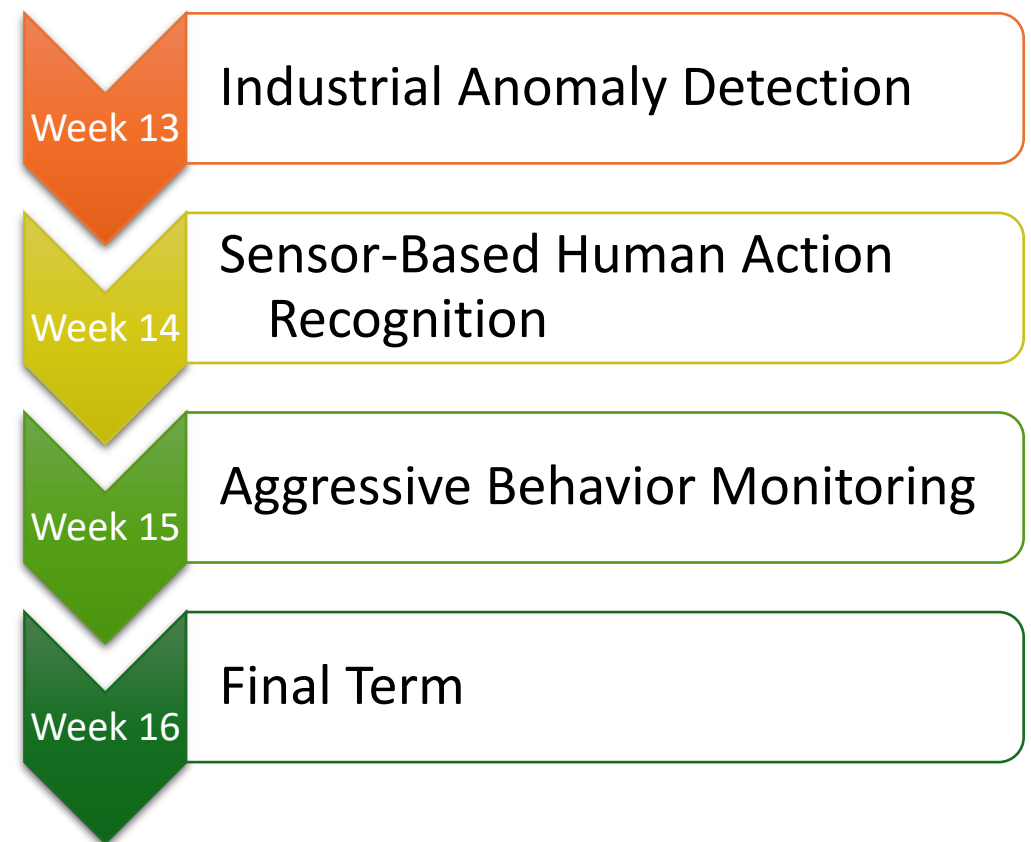


9 (xi): Course Roadmap

Phase 3 Intelligent Multimodal Systems



Phase 4 Beyond Vision



9 (xi): Course Roadmap

1. Professor's Topics of Research

2. Keynote Talk

- Keynote Speaker, The 25th International Conference on Computational Science and Its Applications, Turkey, 2025

3. Intelligent Fire Scene Analysis

- Exposure of Area
- Thesis Tips

4. Students' Presentation

- Topic and Contributions
- Plan and Suggestions

5. Invited Speakers

- Success Story and Experiences
- Projects
- Opportunities

10: Goals and Expectations

1. My goals

- Maximize the learning effectiveness of your time
- Provide roadmap for further exposure to related research areas
- Ensure good grade with hard work, flexibility, and equality

2. My expectations from you

- Attend and participate, when possible
- Follow up the lecture (if missed)
- Start assignments/homework/project etc. well before the deadline
- Stay tuned on the i-Campus
- Feedback about what's working for you and suggest improvements
 - You can write in an email
 - You can submit anonymously
 - Use official week 1, mid-term and final-term evaluations

11: Assignment/Project/Presentation

Dataset Creation and Benchmarking

Instructions

- Total Marks: 15~30
- Each student will do the assignment alone/group.
- Due date for dataset theme selection to avoid duplication (use khanmuhammad@g.skku.edu as usual): **X 2026**
- Due date for submission of all details of the assignment/project/presentation: **XX 2026**

Task: Each student/team is required to make a comprehensive plan of a new dataset, including its title, main description of the dataset, number of classes, images count and a table listing existing datasets with their key information and details of the new dataset, showing how the new dataset is different and unique etc. You can include all possible details. I will explain via example in the class or using a separate video. The evaluation will be done based on these points:

1. Dataset Understanding & Problem Analysis (5~10 points)
2. Novel Dataset / Extension Proposal (5~10 points)
3. Practical Demonstration/Feasibility (5~10 points)
 - **Option A (Coding track):** run a prototype experiment, test a model, do possible ablation and report results.
 - **Option B (Non-coding track):** provide a feasibility analysis, workflow diagram, collection plan, and manual annotations sample for the dataset you are proposing.
 - **Option C: Create a small/mini dataset (what is proposed in point 2)**

All details can be finalized once the final numbers of students are confirmed.

Thank You

Any Question?

Feel free to contact on

khanmuhammad@g.skku.edu

Welcome to Join!